Iran Renewable System Analysis

Group A | February 5th, 2026

Data Science for Energy System Modelling

Outline

Introduction

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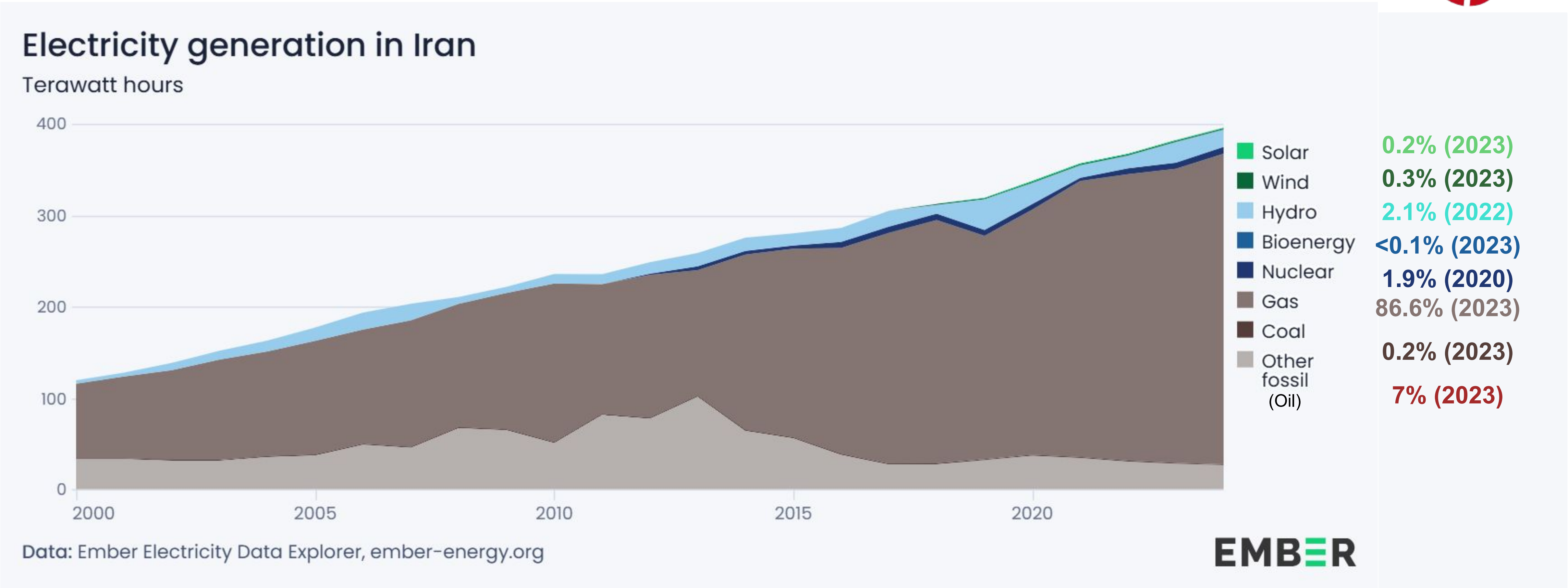
Context - Current State of Electricity System in Iran



Fact Sources: **(IR Gas Production, 2024), (IR Oil Production, 2023)**

Context - Model Considerations

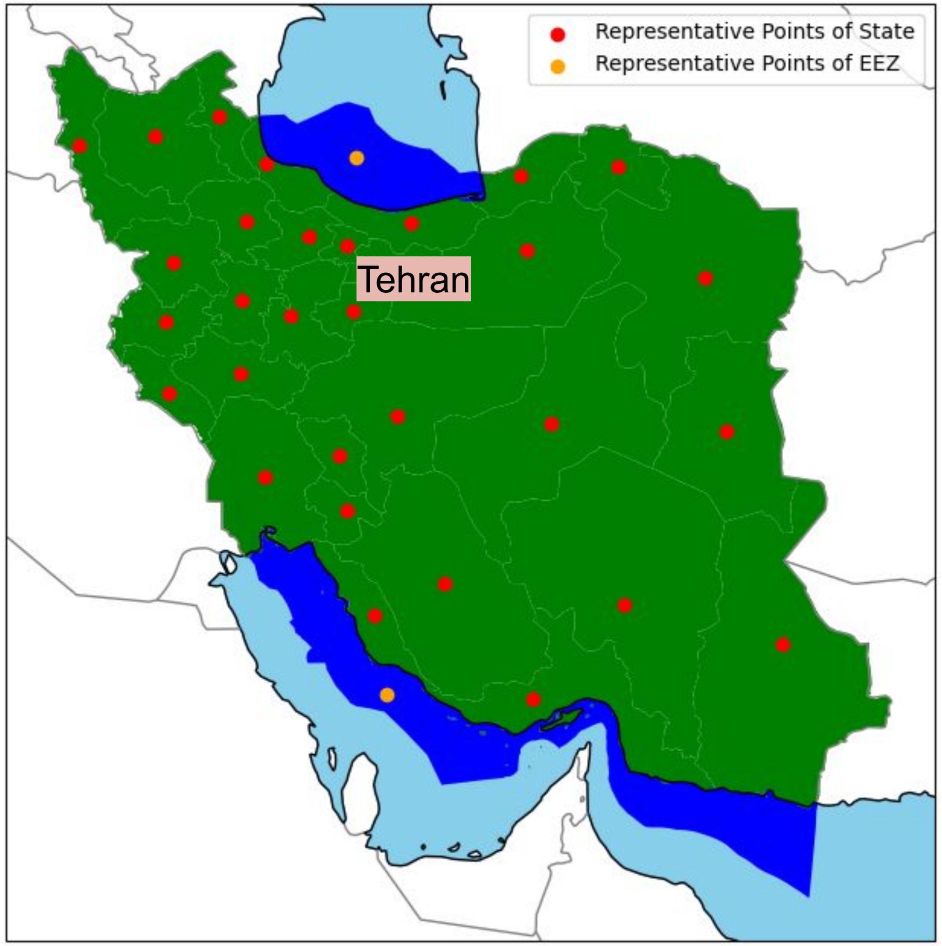
- Area: 1.648 million km²
  (vs. Germany: 0.357 million km²)

- Population (2024): ~85,961,000
  (vs. Germany ~83,600,000)

- System costs sourced from various currency years
- Cost projection year: 2025
- Discount Rate: 7%

- 31 States + 2 exclusive economic zones (EEZ)
- 1 Bus per state/EEZ
- Buses set at regional representative point

Sources: **(Bundesamt, 2024)**, **(Statistical Center of Iran, 2024)**



Model Inputs

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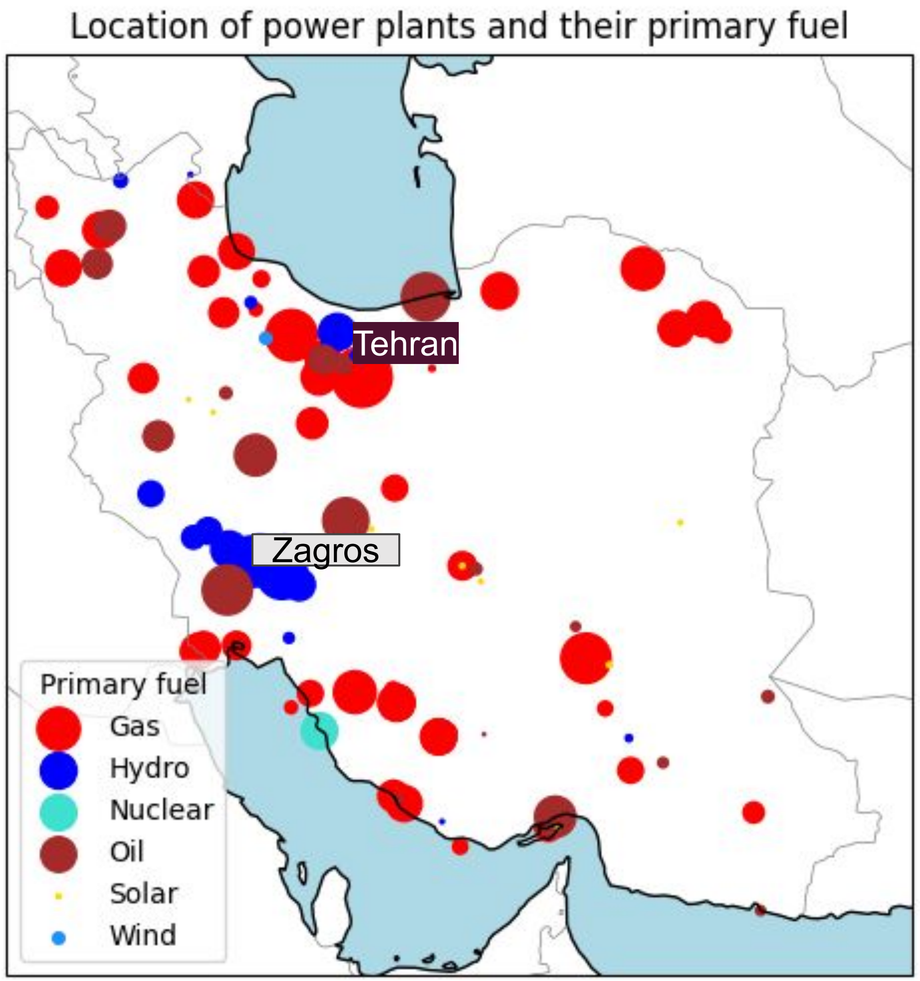
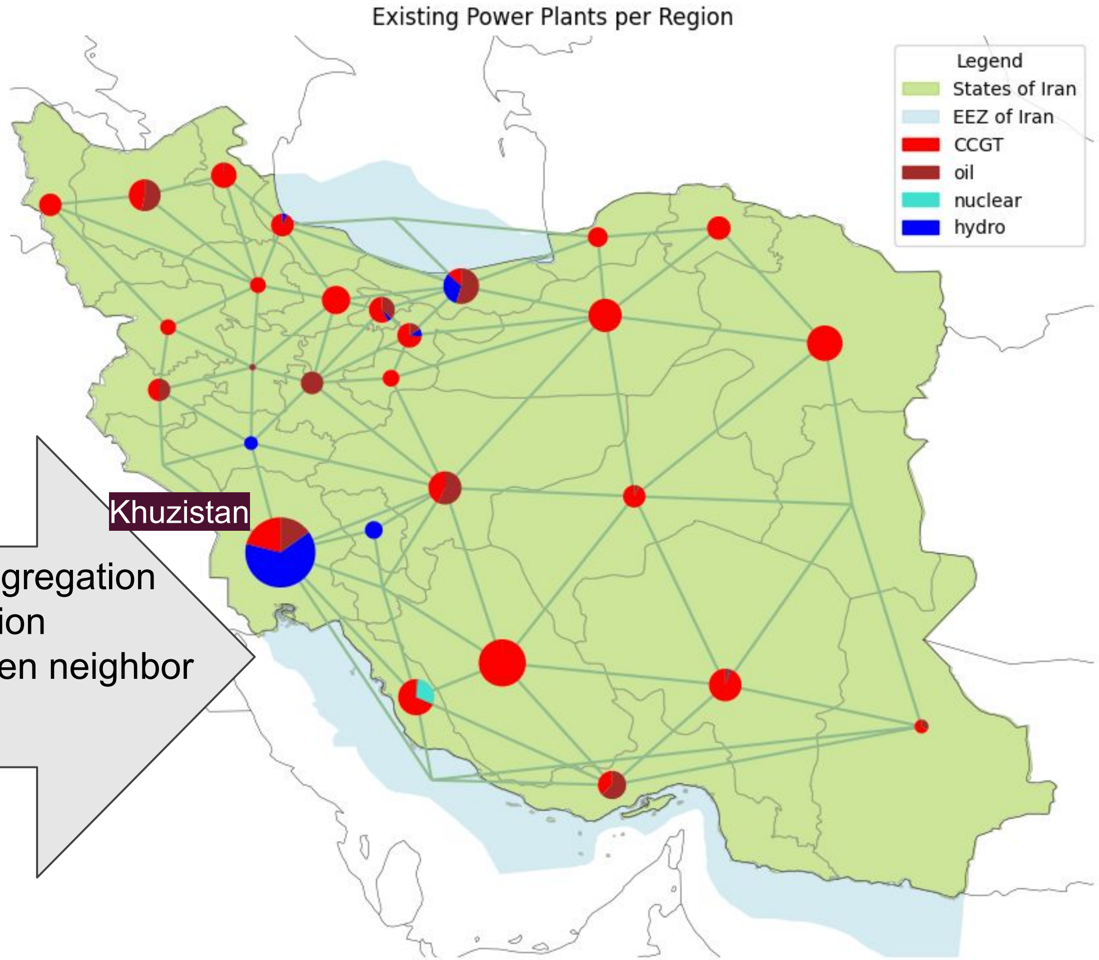
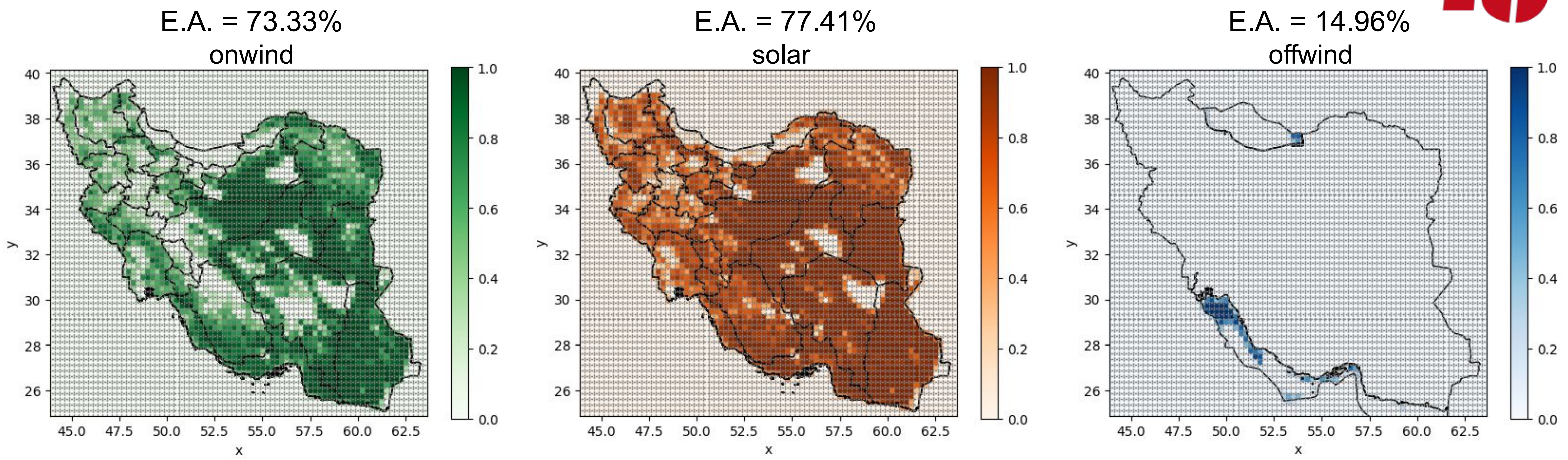
Inputs - Power Plant Capacities



- Regional Aggregation
- RES Exclusion
- Links between neighbor regions

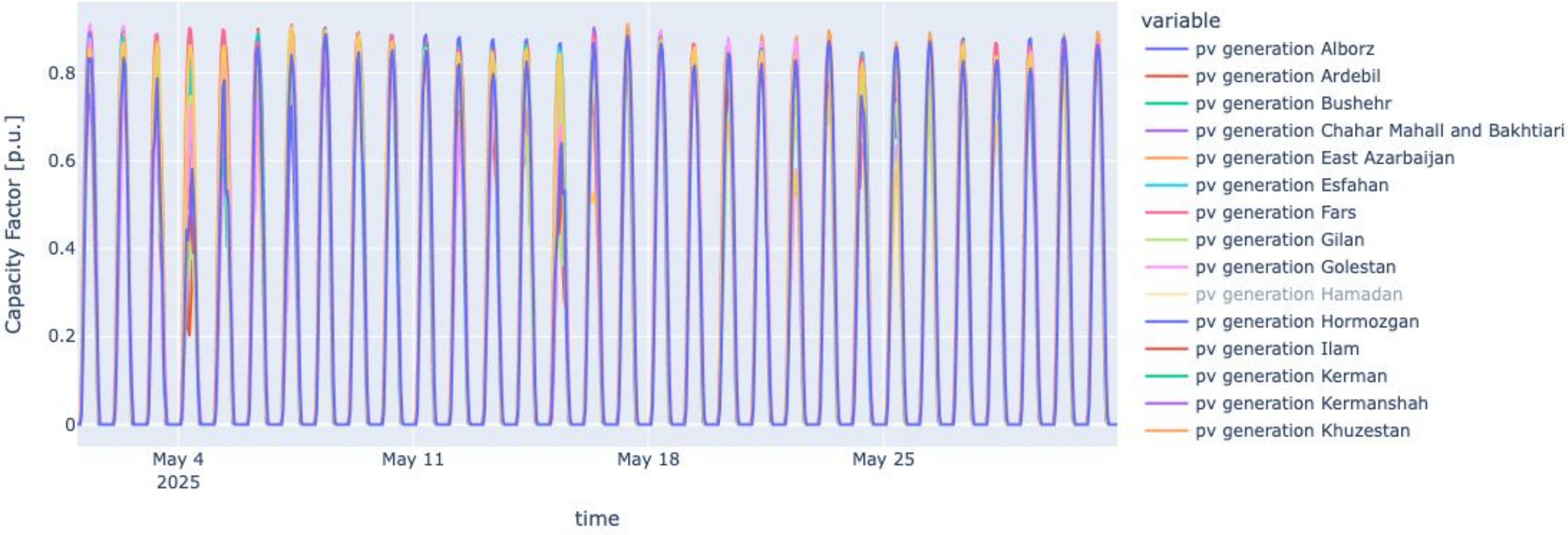


Inputs - Land Eligibility Analysis

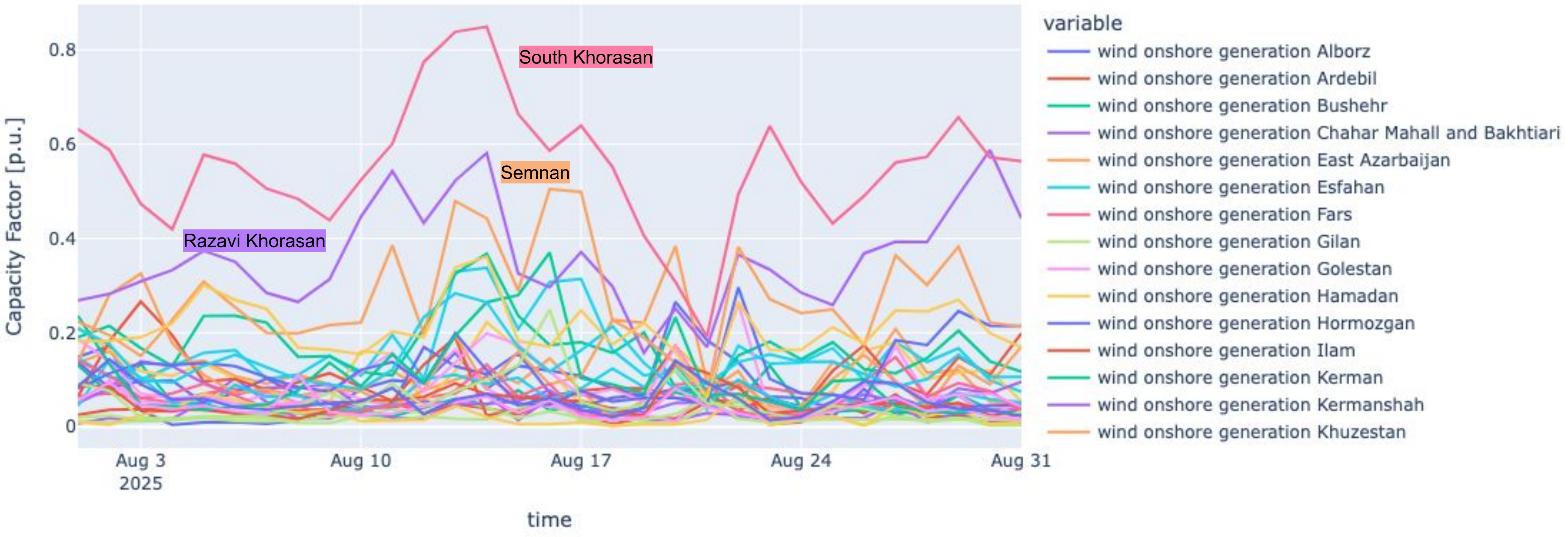


- Considered criteria:
  - Elevation data
  - Distance from airports, major roads, built up areas
  - Exclusion of world protected areas
  - Exclusive Economic Zones (for wind offshore)
  - Land classification

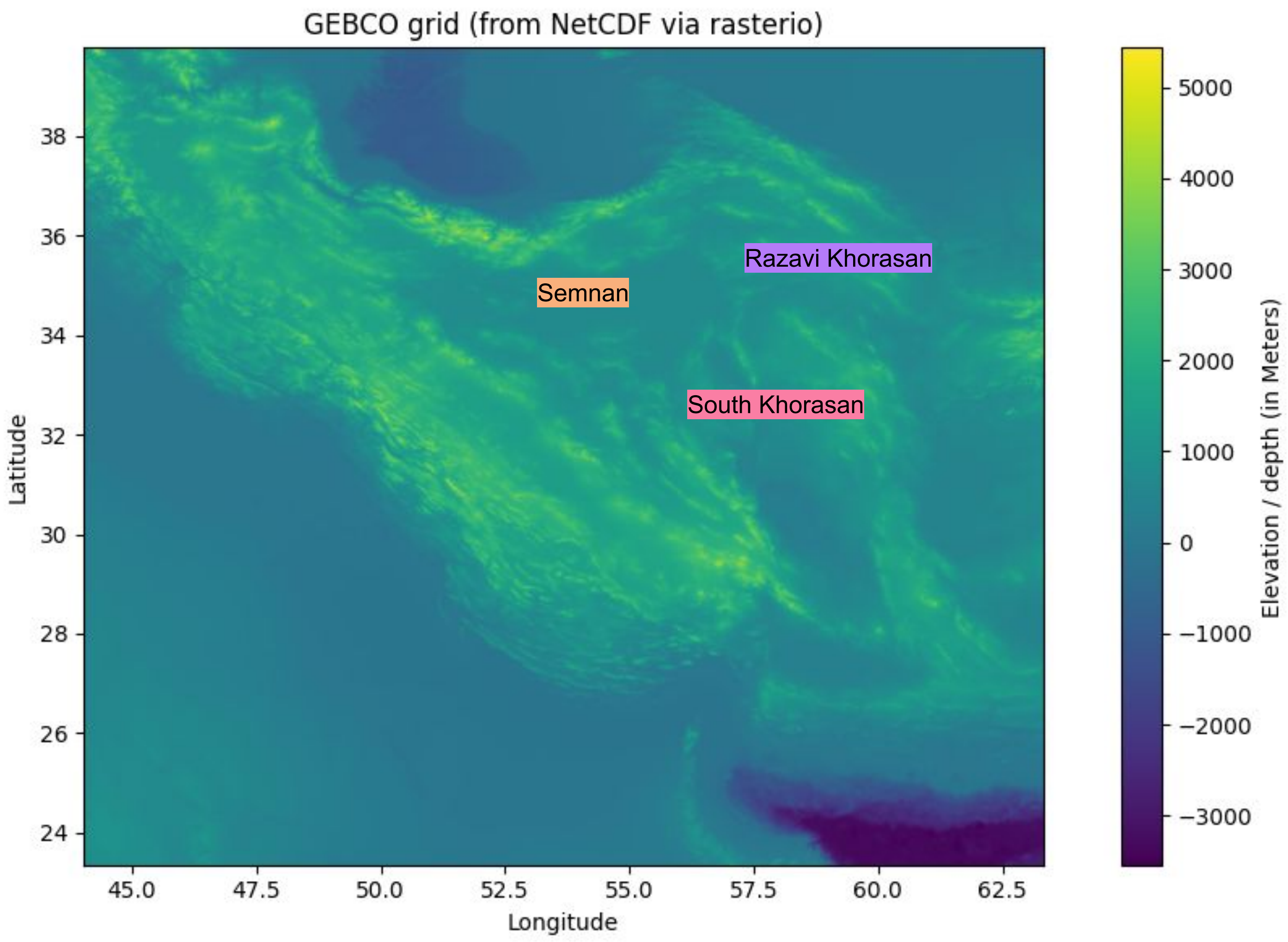
Inputs - RES Time Series (PV - May - 2025)



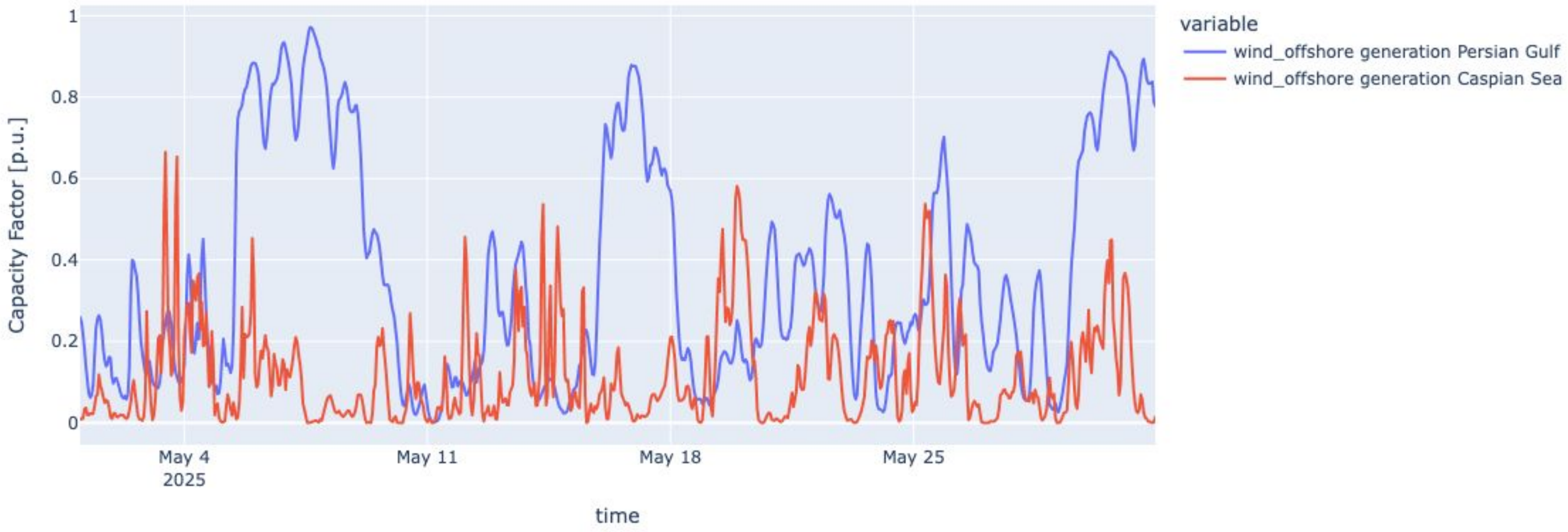
Inputs - RES Time Series (Wind onshore - August 2025)



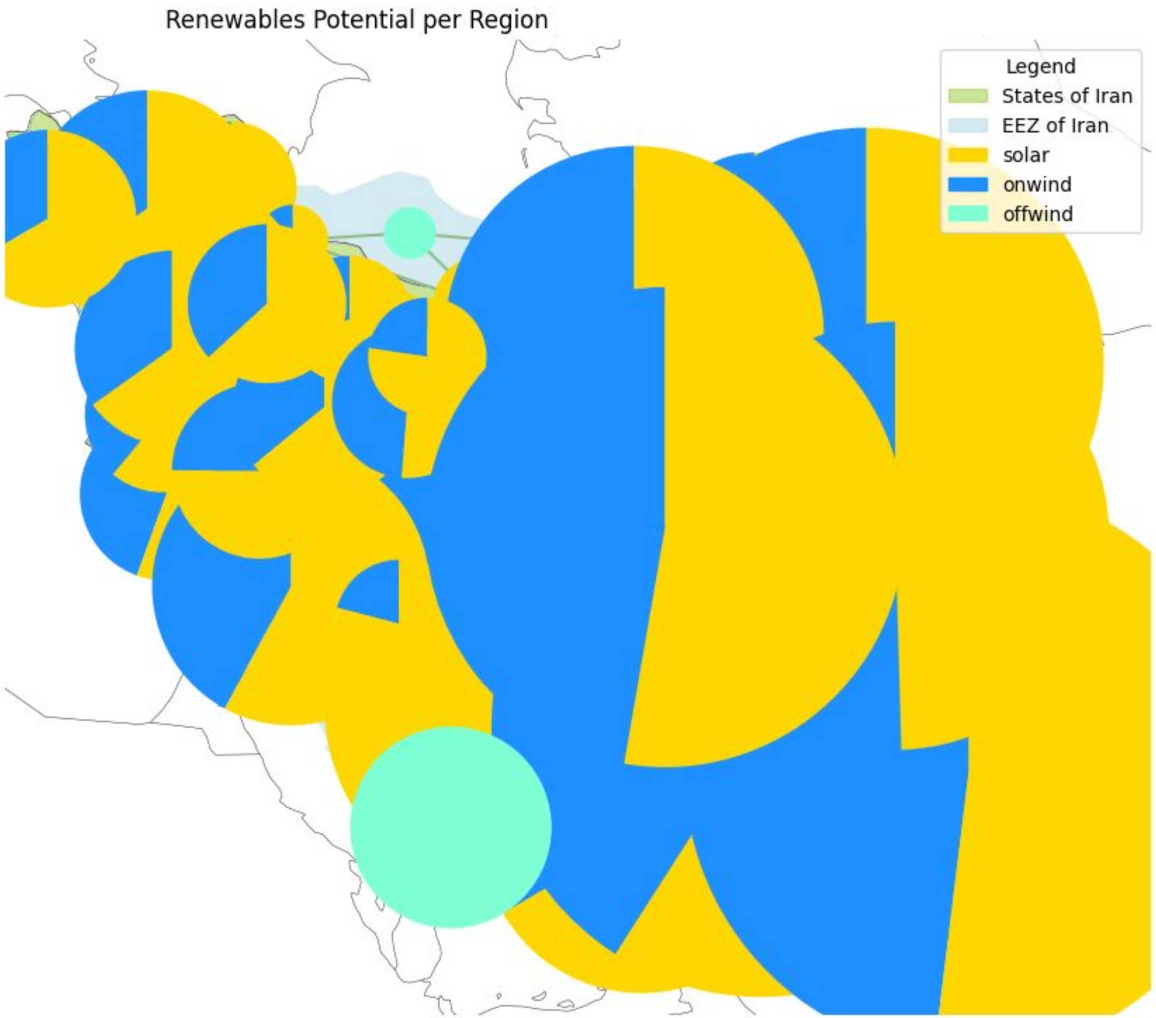
Inputs - Elevation Data

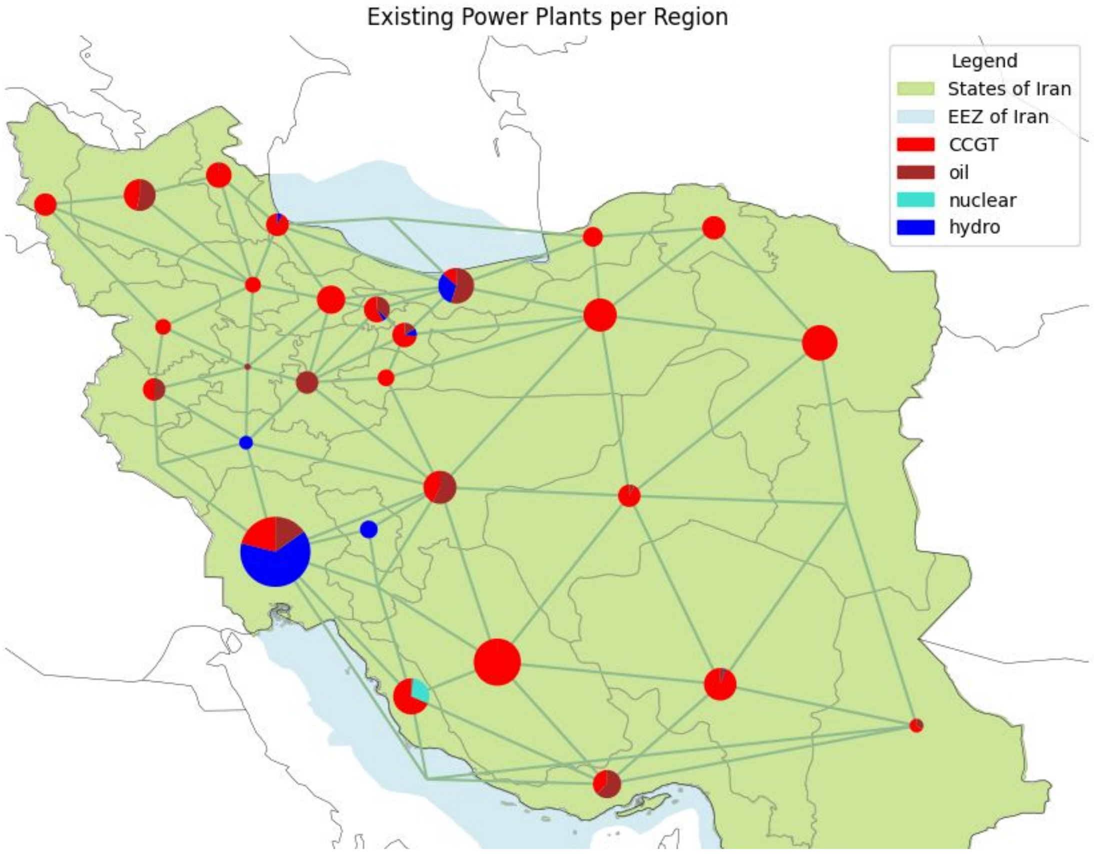


Inputs - RES Time Series (Wind offshore - May 2025)



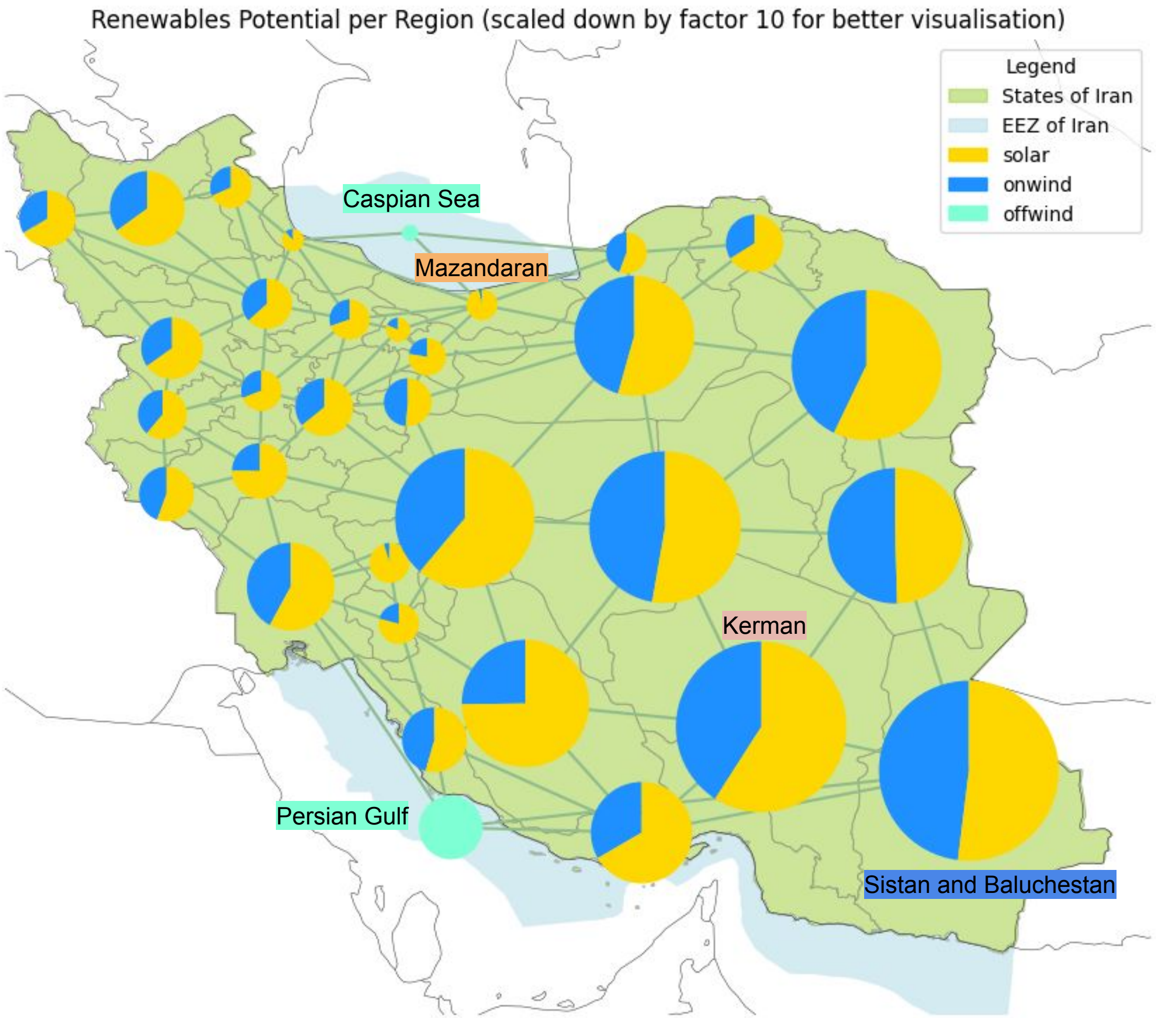
Inputs - RES Available Capacity (same scale of existing power plants)





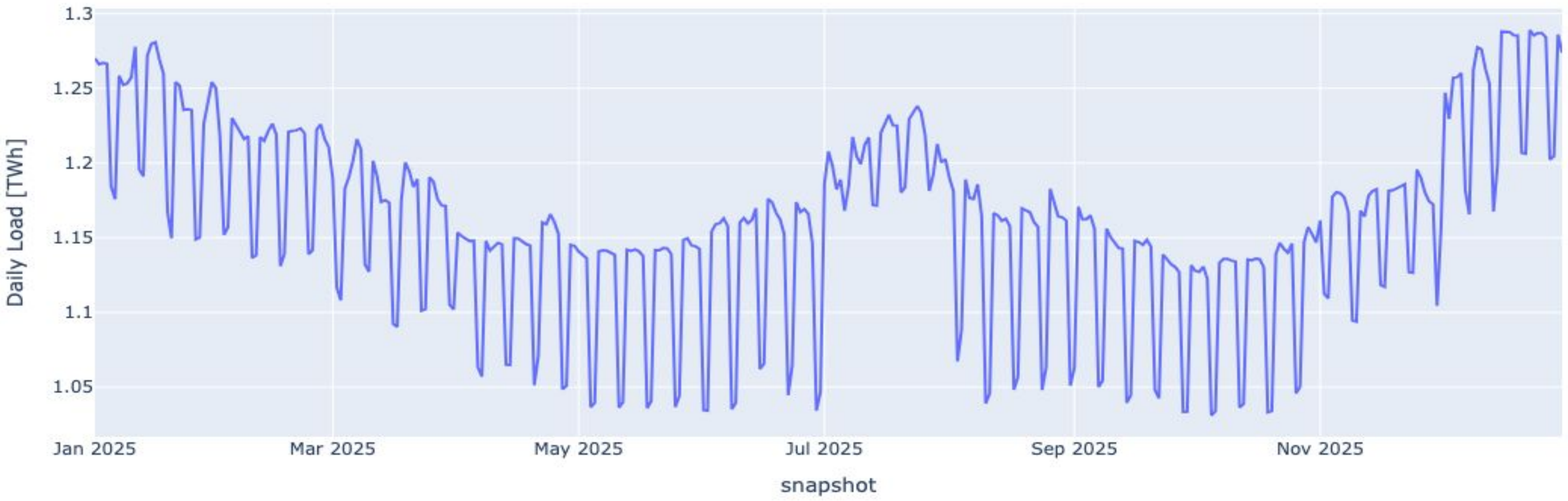
- The potentials are in the same scale as the previous plot, showing existing power plants!

Inputs - RES Available Capacity (scaled down by factor 10)

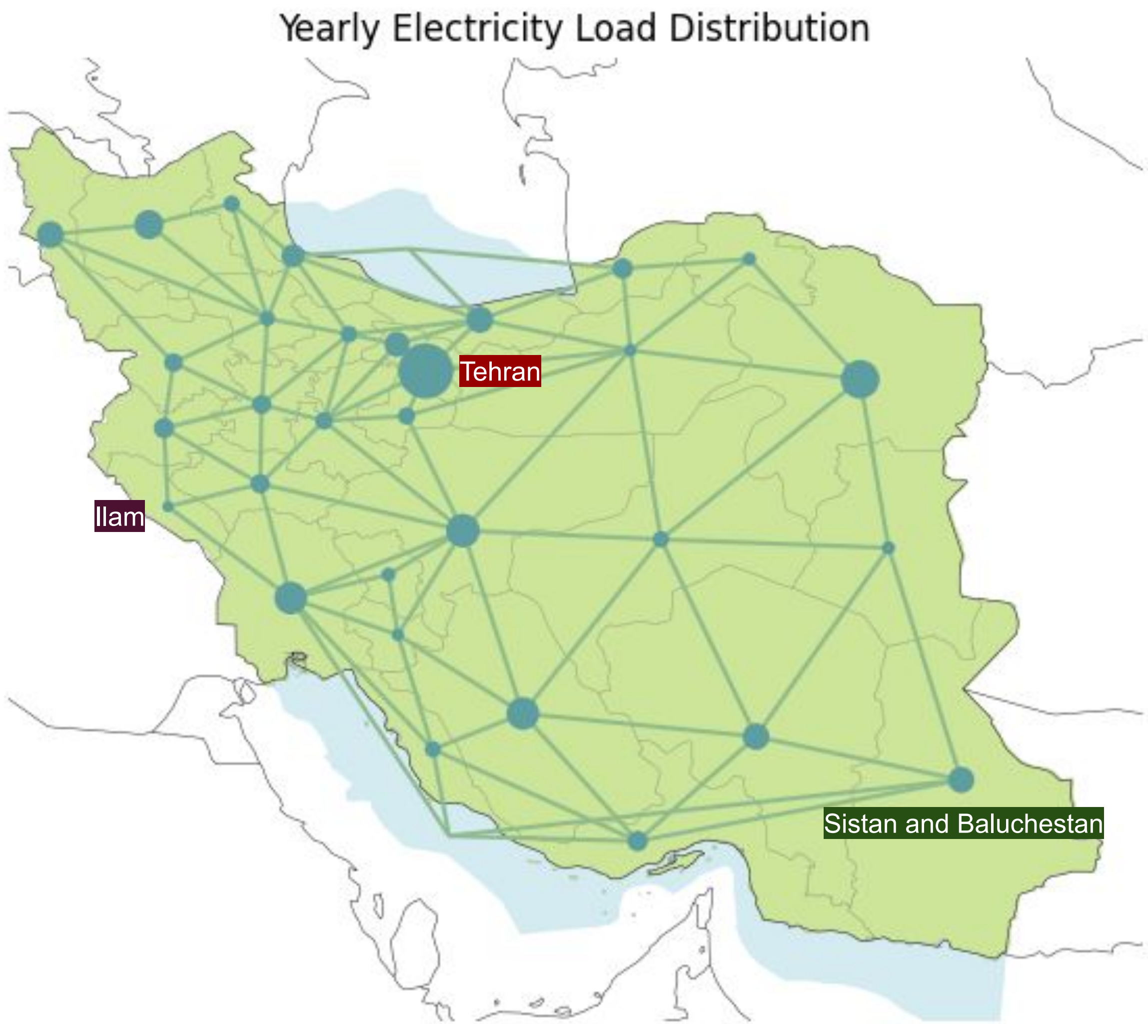


- Base: Land Eligibility Analysis and 2025 weather
- High Potential of Solar and Wind Capacities, especially in the center and east
  - Sistan and Baluchestan:
    - Solar: ~380 GW
    - Wind onshore: ~350 GW
  - Kerman:
    - Solar: ~390 GW
    - Wind onshore: ~270 GW
  - Mazandaran:
    - Solar: ~21 GW
    - Wind onshore: ~0.8 GW
  - Persian Gulf:
    - Wind offshore: ~92 GW
  - Caspian Sea:
    - Wind offshore: ~6 GW

Inputs - Load Time Series



Inputs - Regional Load Distribution



- Load time series (from 2013 projected to 2030)
- Electricity demand of ~424 TWh/a
- Assumption: Distributed to regions, based on population
- Load per Region ranges between:
  - Tehran: ~71 TWh/a
  - …
  - Sistan and Baluchestan: ~16 TWh/a
  - …
  - Ilam: ~3 TWh/a

Model Results

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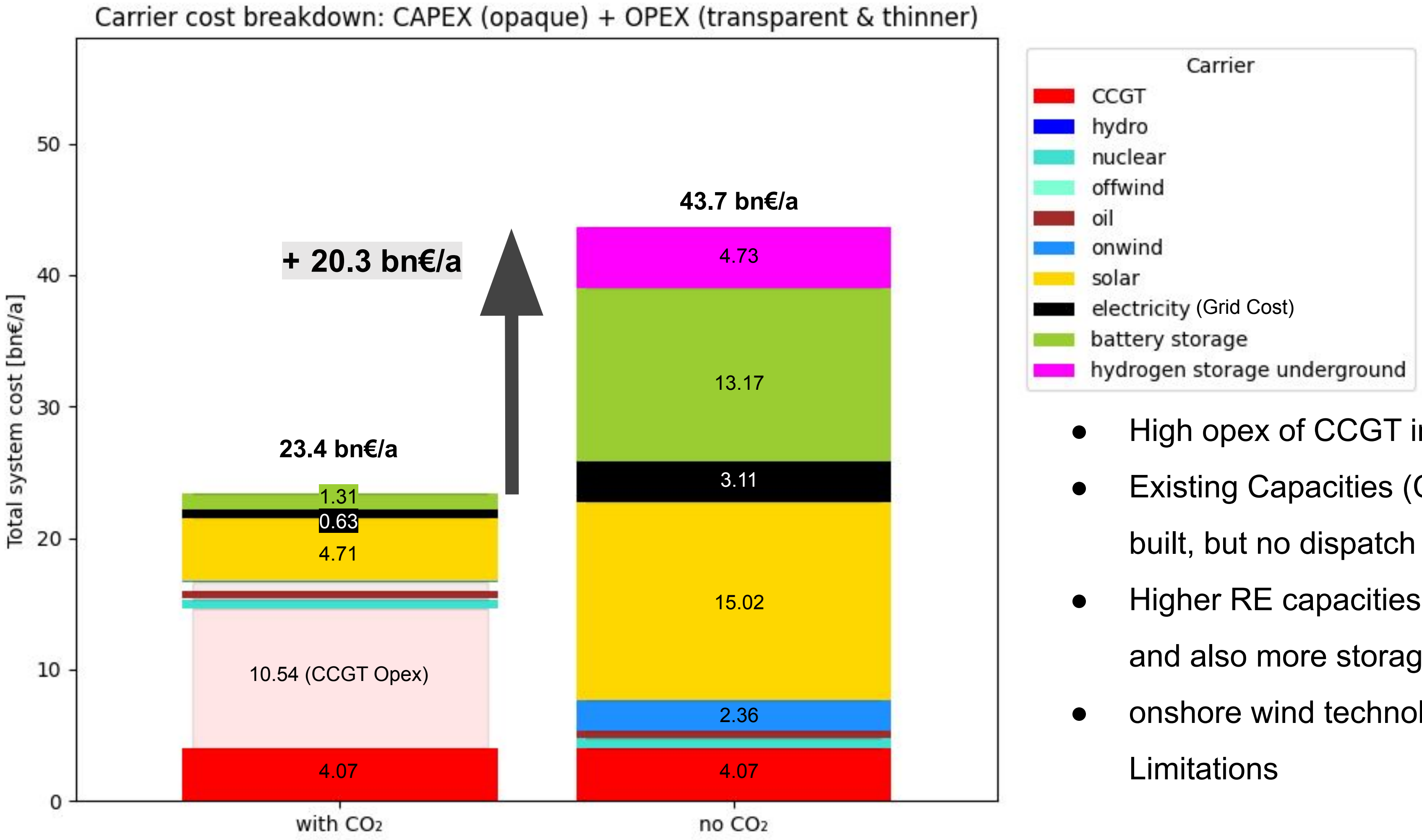
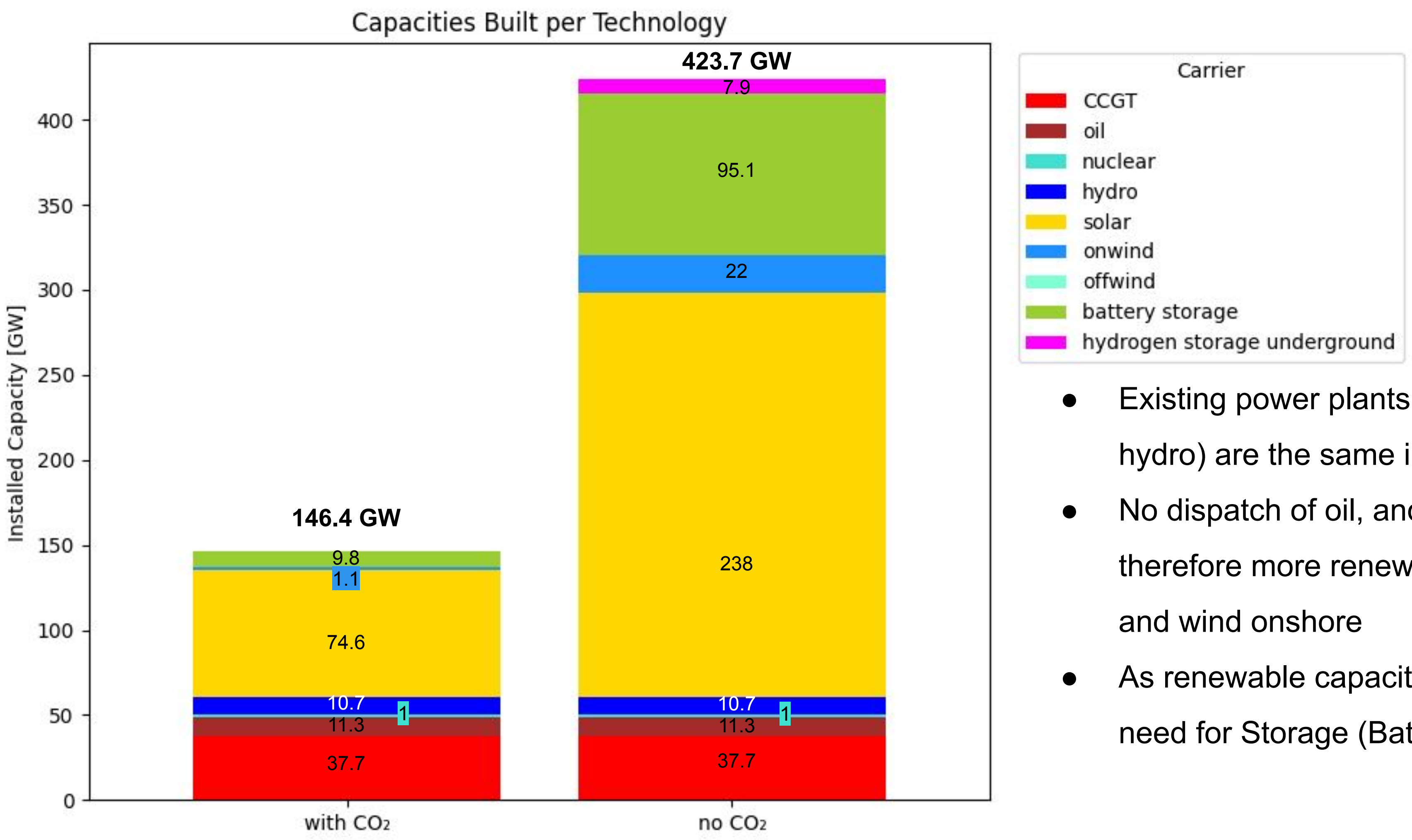
Results - Total System Costs



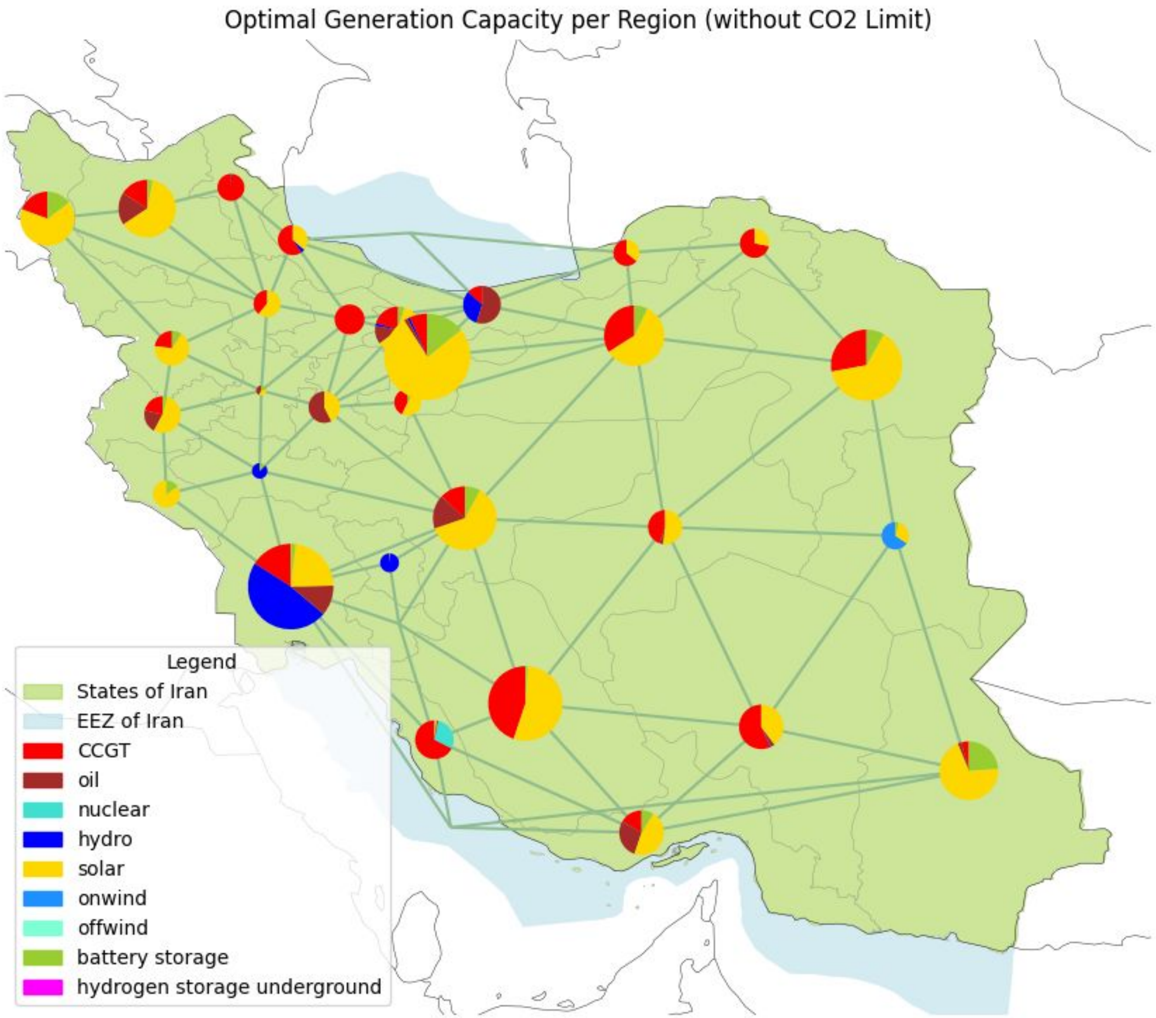
- High opex of CCGT in base case (with CO2)
- Existing Capacities (CCGT, Oil, Nuclear) always built, but no dispatch of CCGT and Oil in “No CO2”
- Higher RE capacities require more grid investment, and also more storages
- onshore wind technology only required with CO2 Limitations

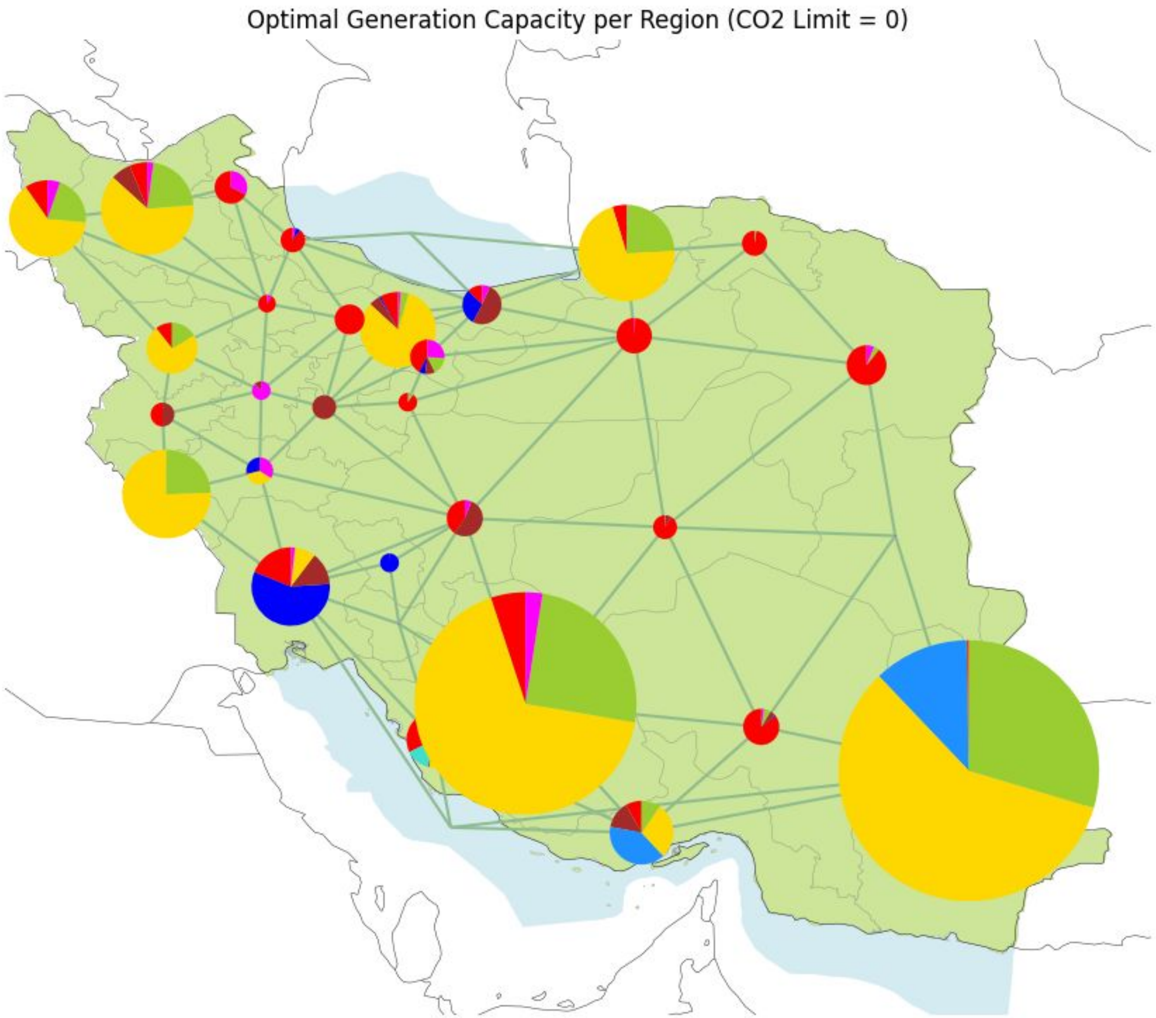
Results - Optimal Generation Capacity



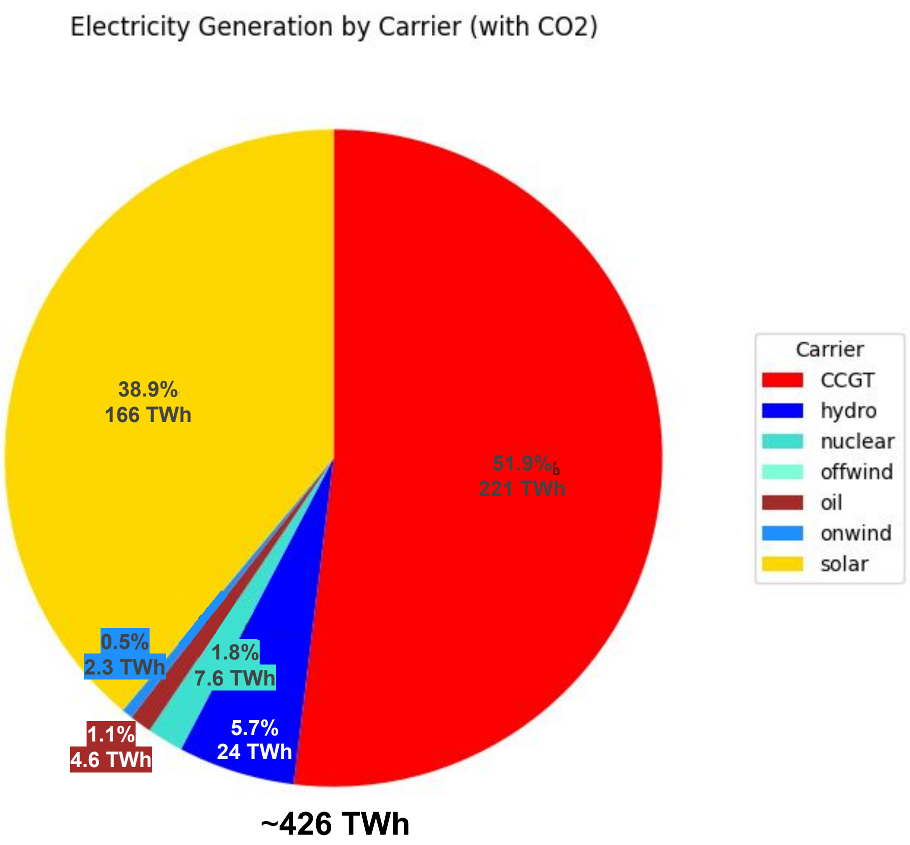
- Existing power plants (CCGT, oil, nuclear and hydro) are the same in both cases
- No dispatch of oil, and CCGT in “no CO2” case, therefore more renewable capacities, mostly solar and wind onshore
- As renewable capacity increases, there is more need for Storage (Battery and Hydrogen)

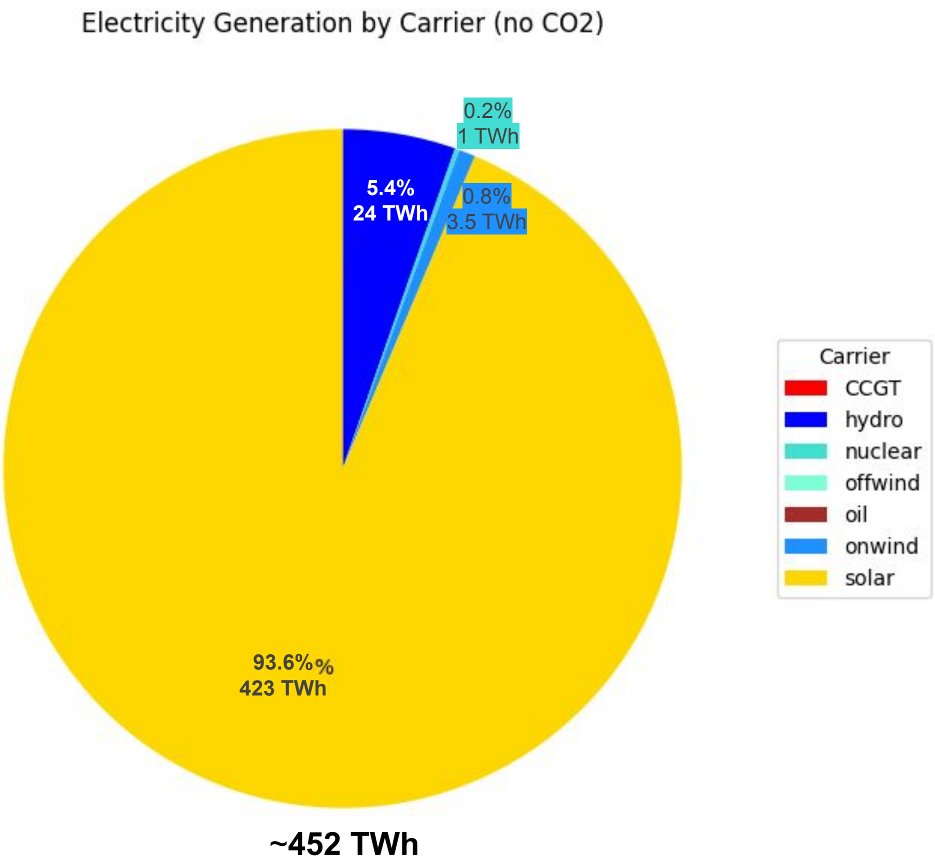
Results - Optimal Generation Capacity per Region





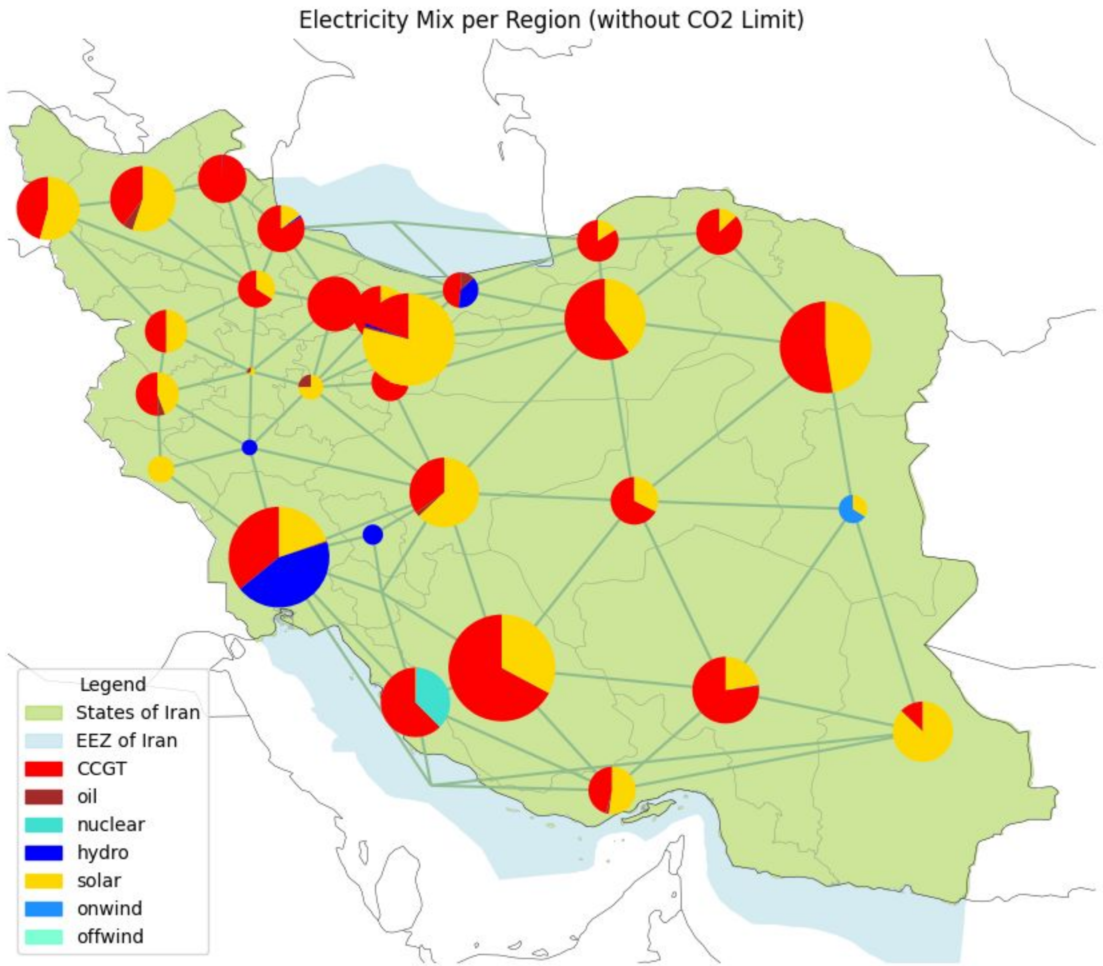
Results - Electricity Mix

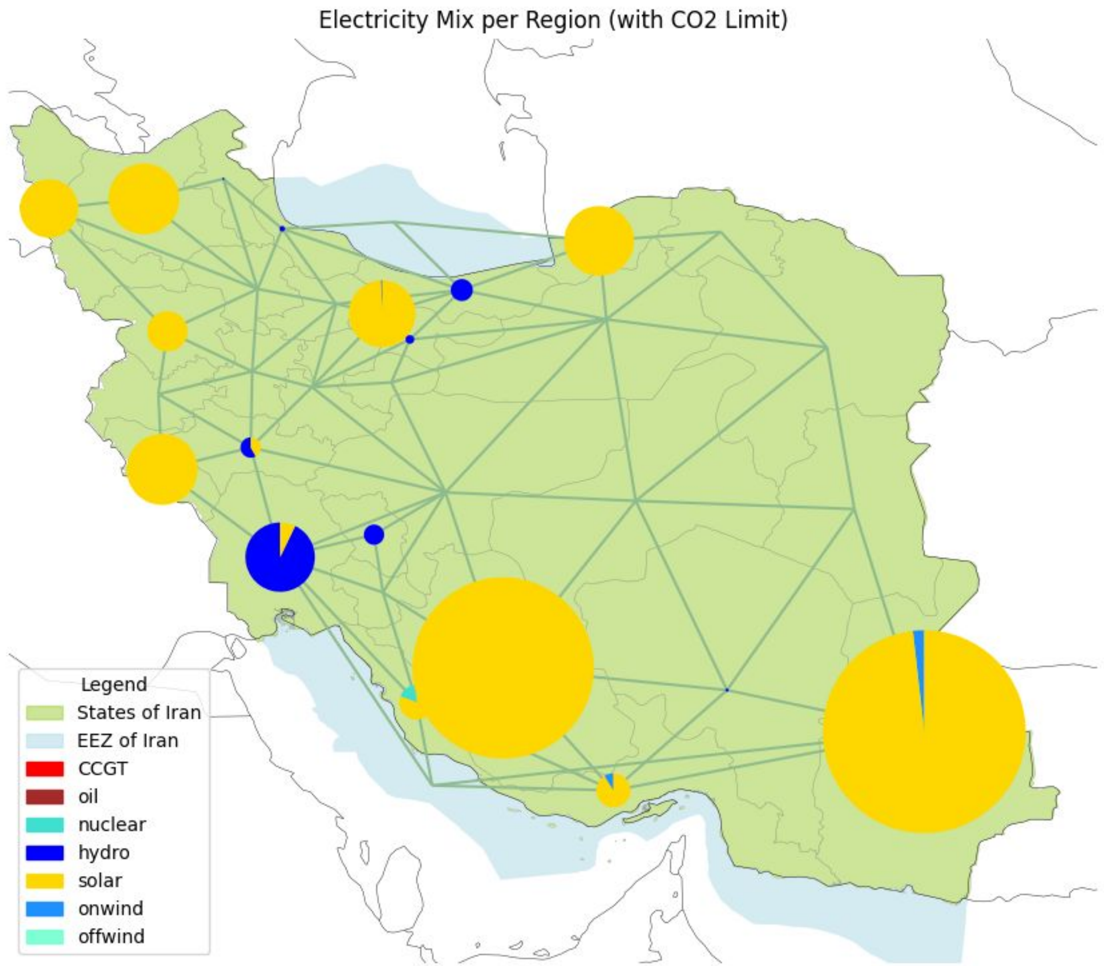




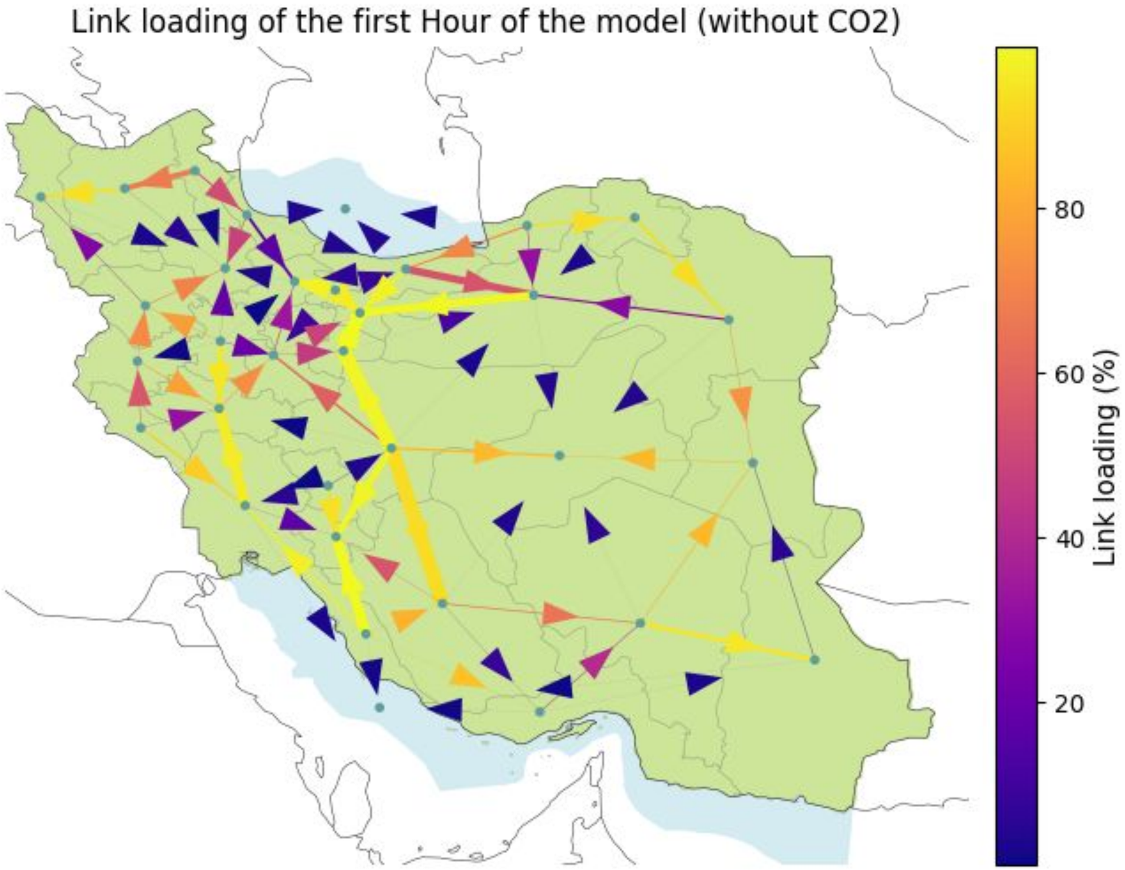
- More generation than load mostly because of the losses in Storages

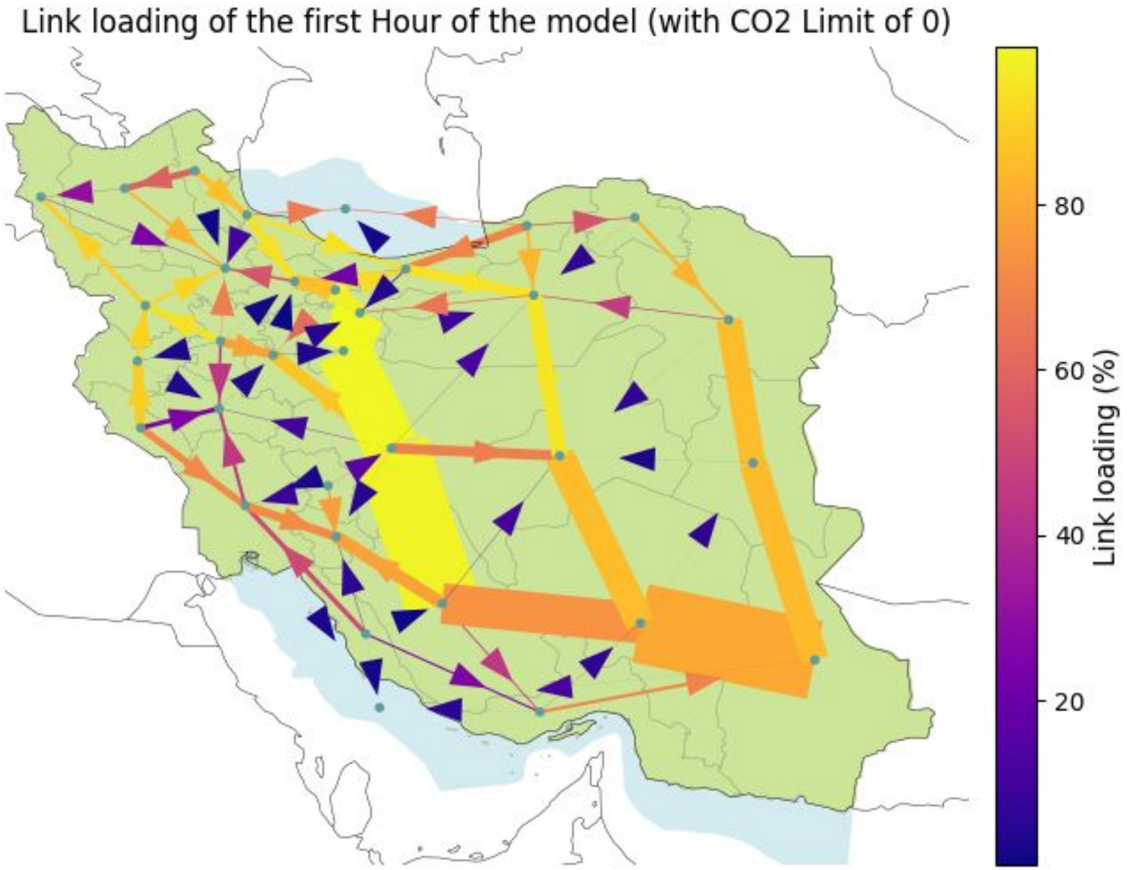
Results - Electricity Generation Mix per Region



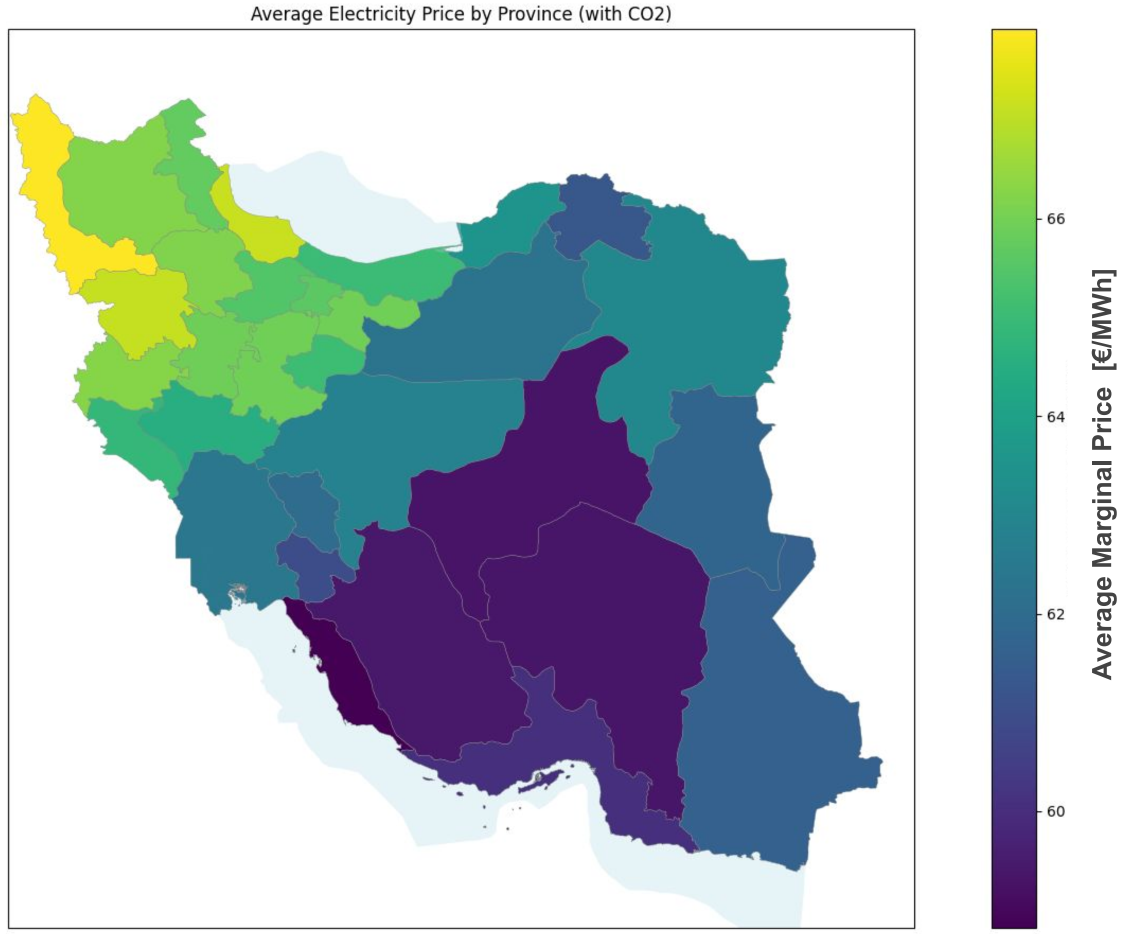


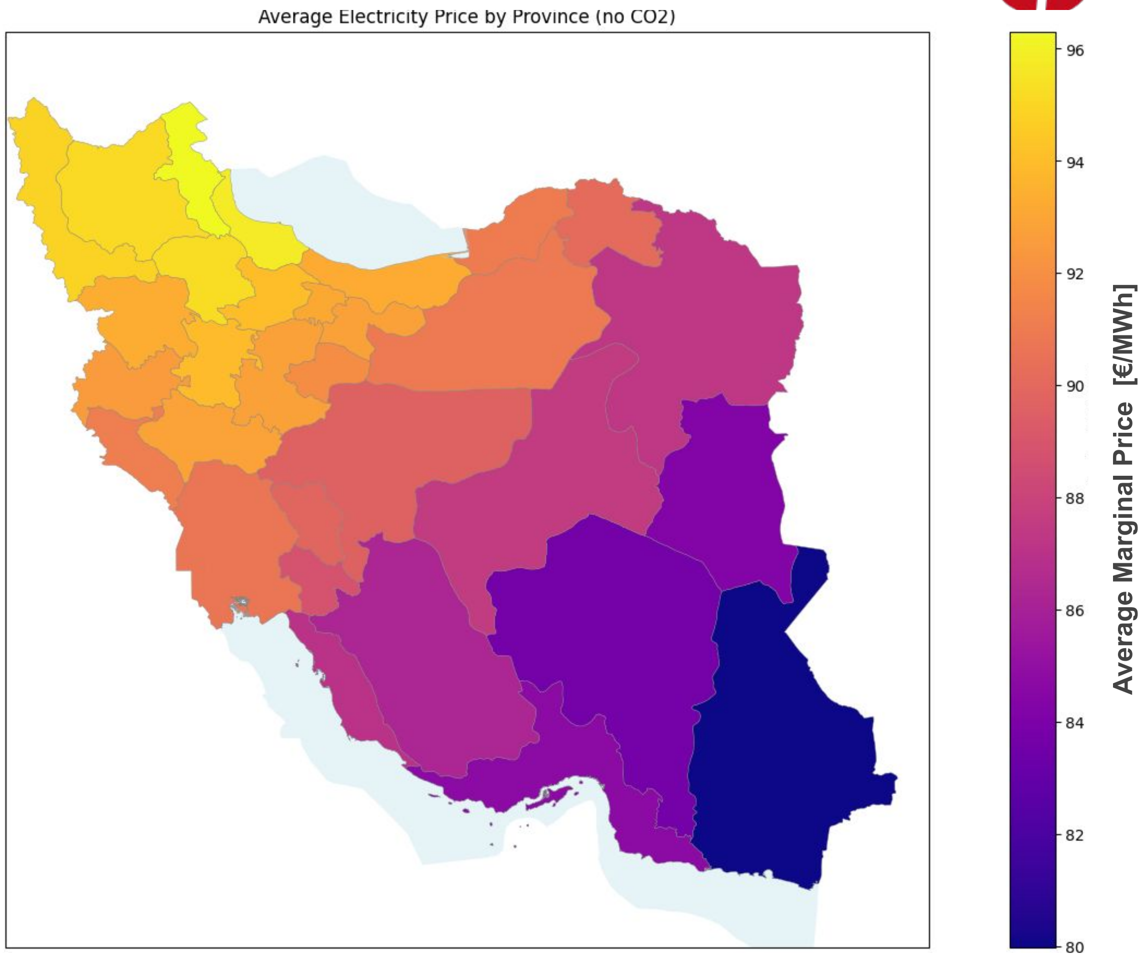
Results - Power Flow



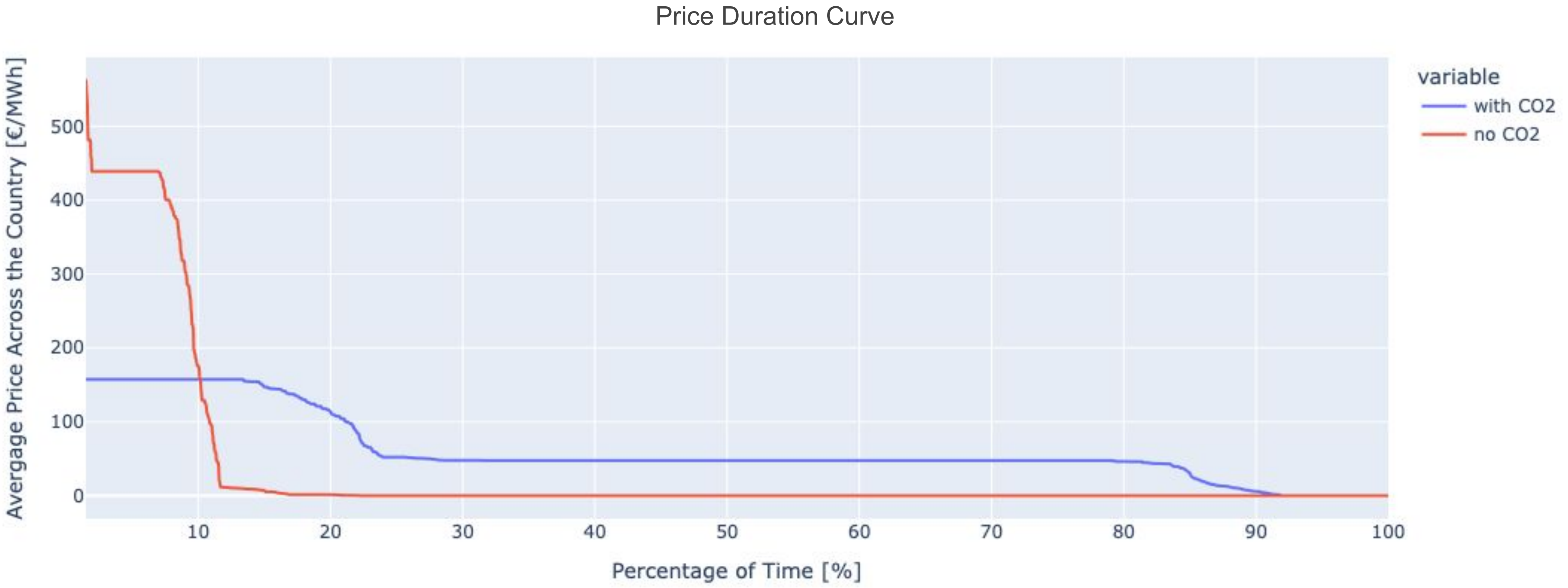


Results - Average Price per Region



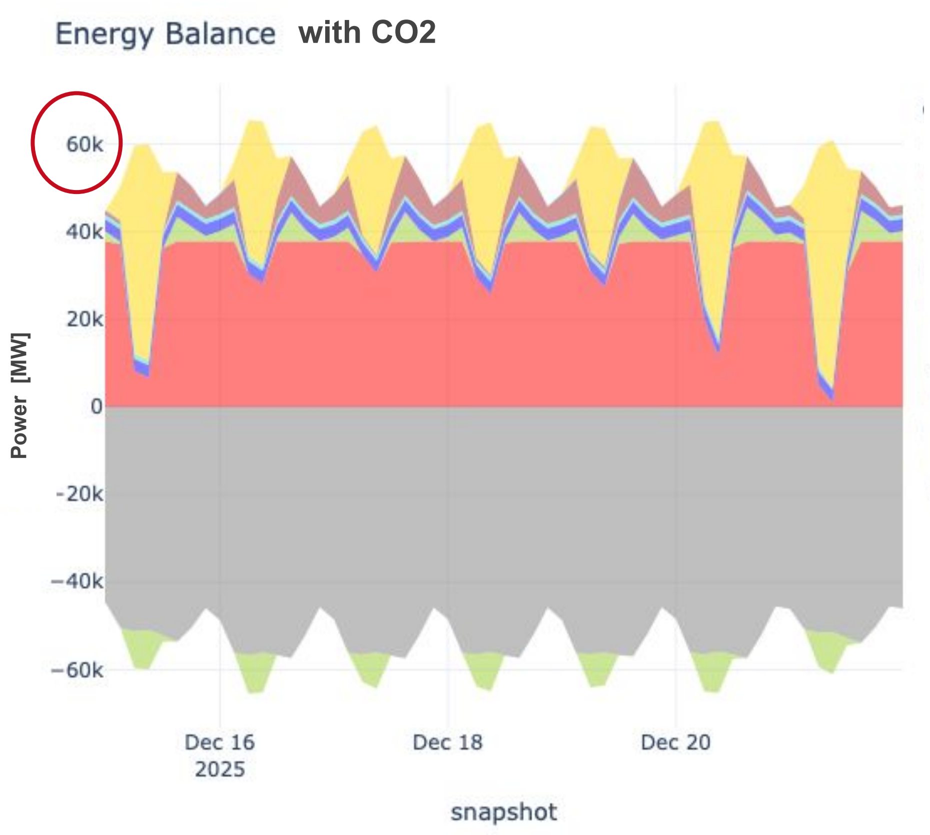


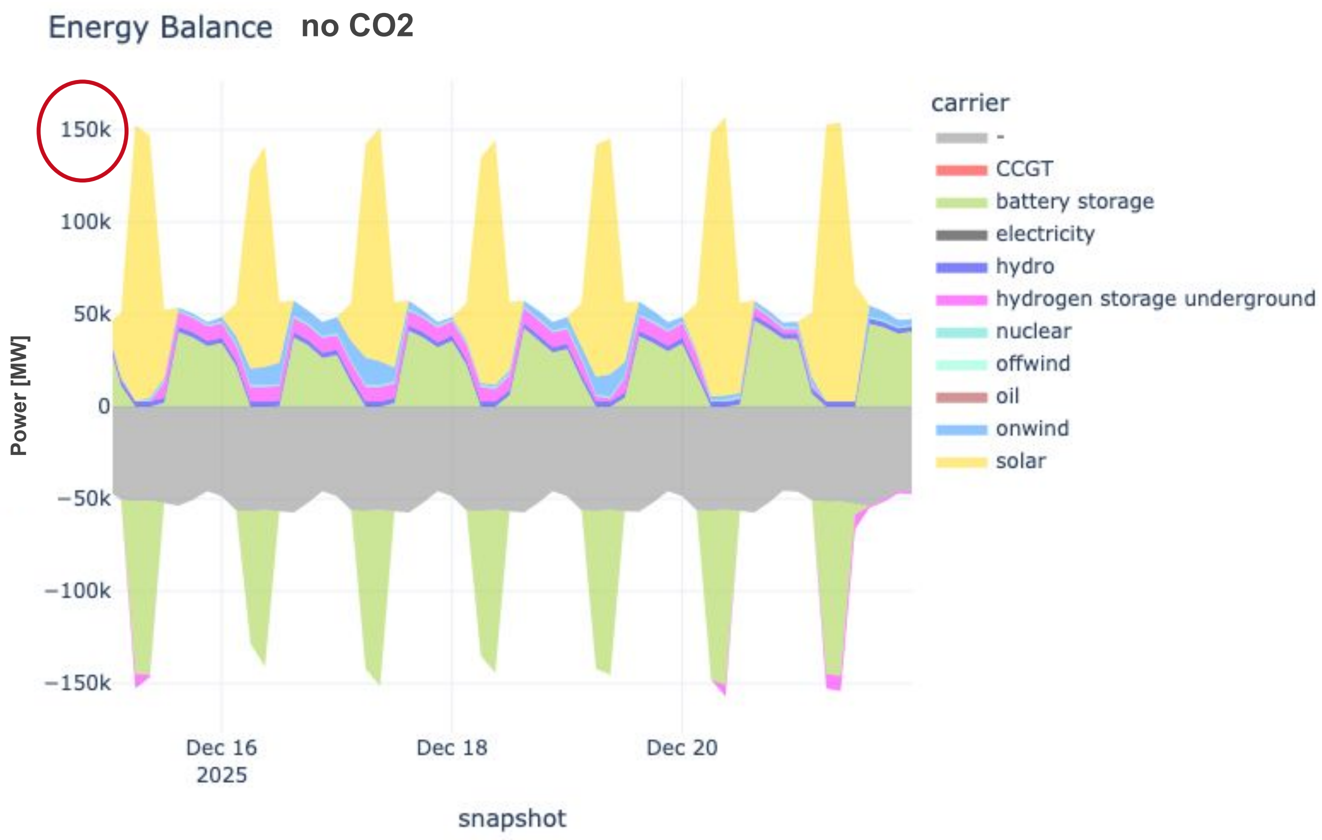
Results - Prices



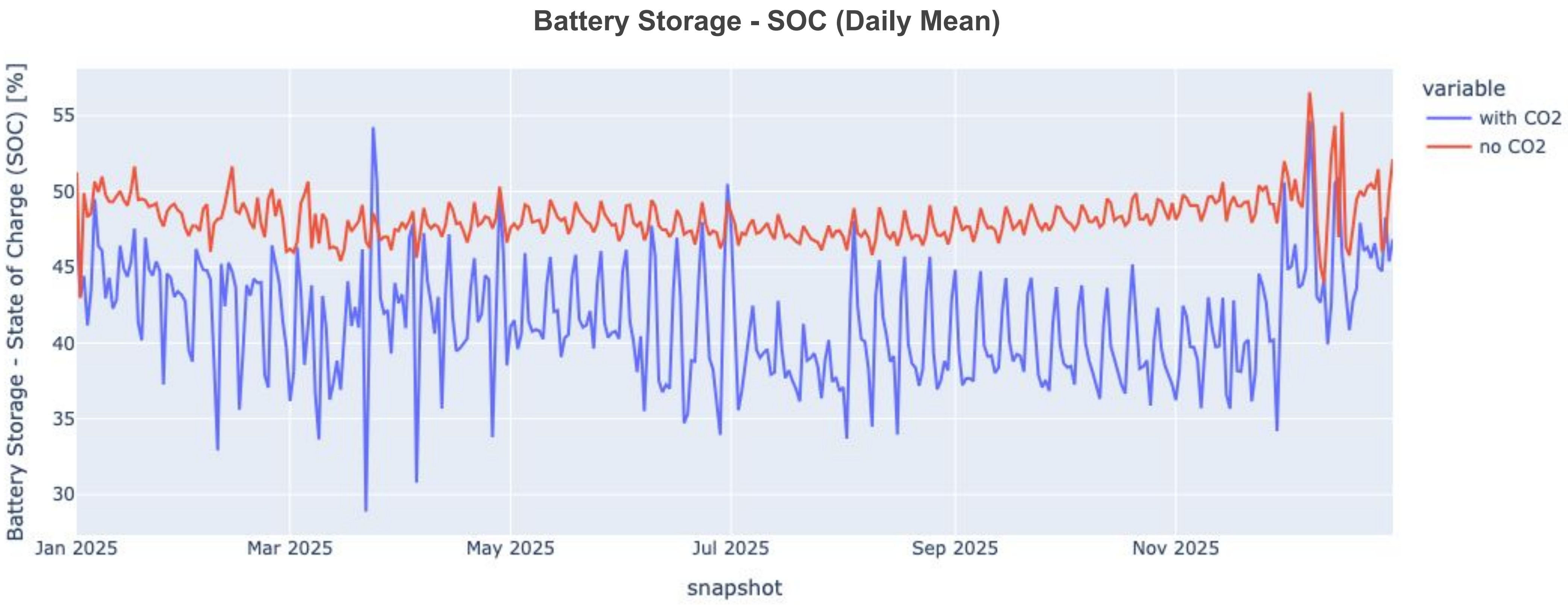
Note: the top 1.5% percent are removed for better visualization (see archive for more detail)

Results - Energy Balance



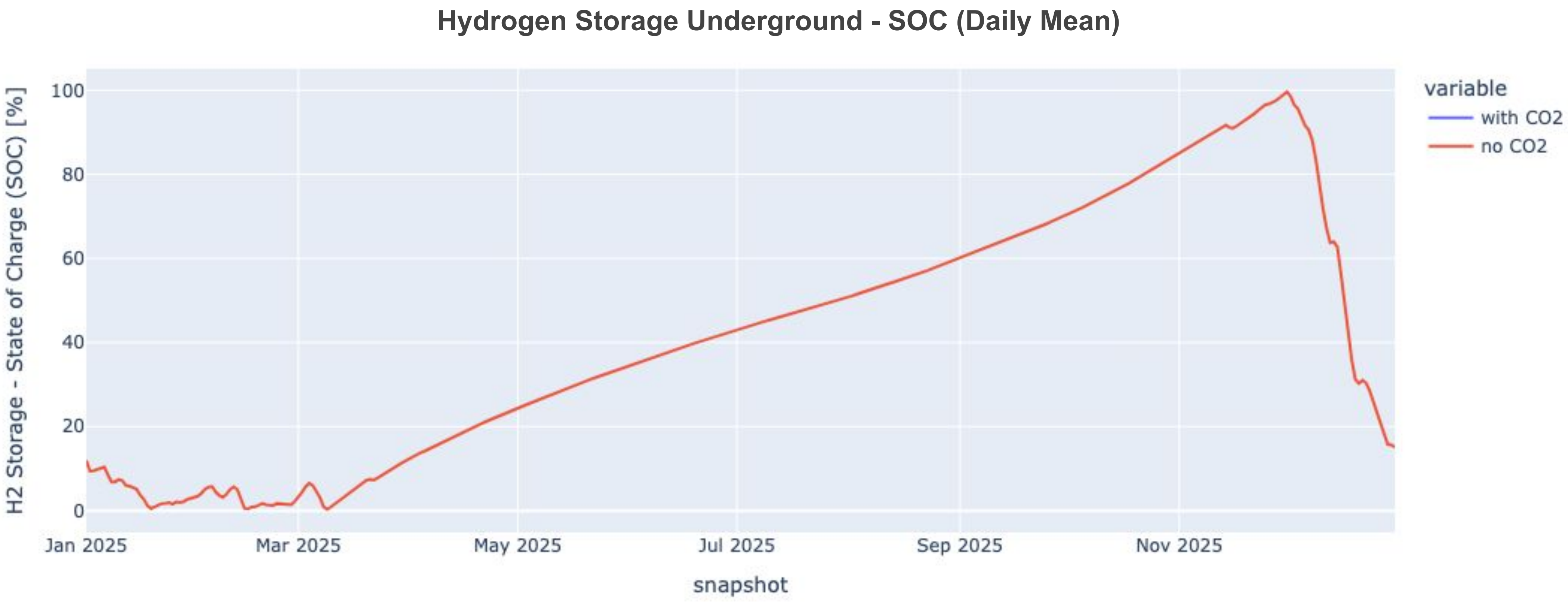


Results - Storages



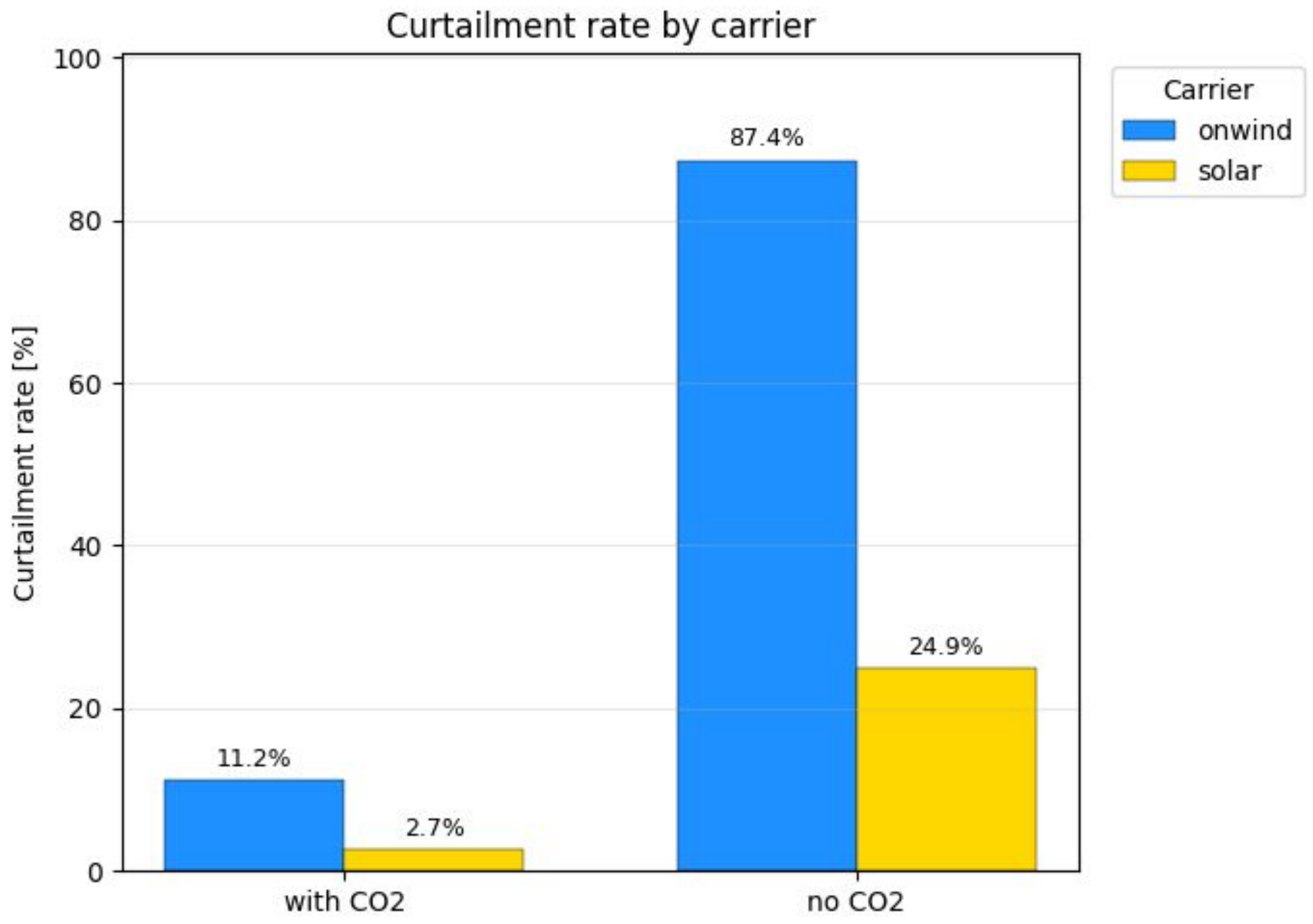
100% SOC: **55.3 GWh (with CO2)** vs. **562.9 GWh (no CO2)**

Results - Storages



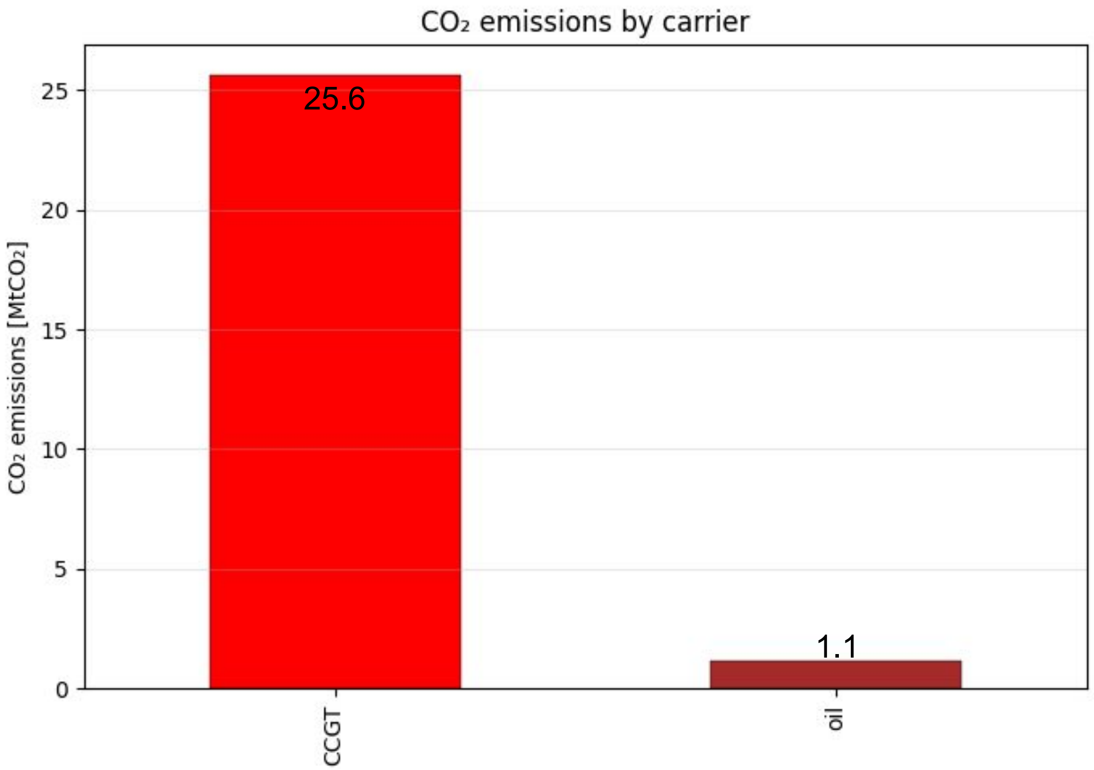
100% SOC: **5328.6 GWh (no CO2)**

Results - Rate of Curtailment



Results - CO2 Shadow Prices

with CO2 Case:



no CO2 Case:

Total emissions:	0.0	tCO2
Carbon shadow price:	15,389.3	€/tCO2

→ Allowing 1 extra tonne of CO2 would reduce total system cost by ~15,000 €

Sensitivity Analysis

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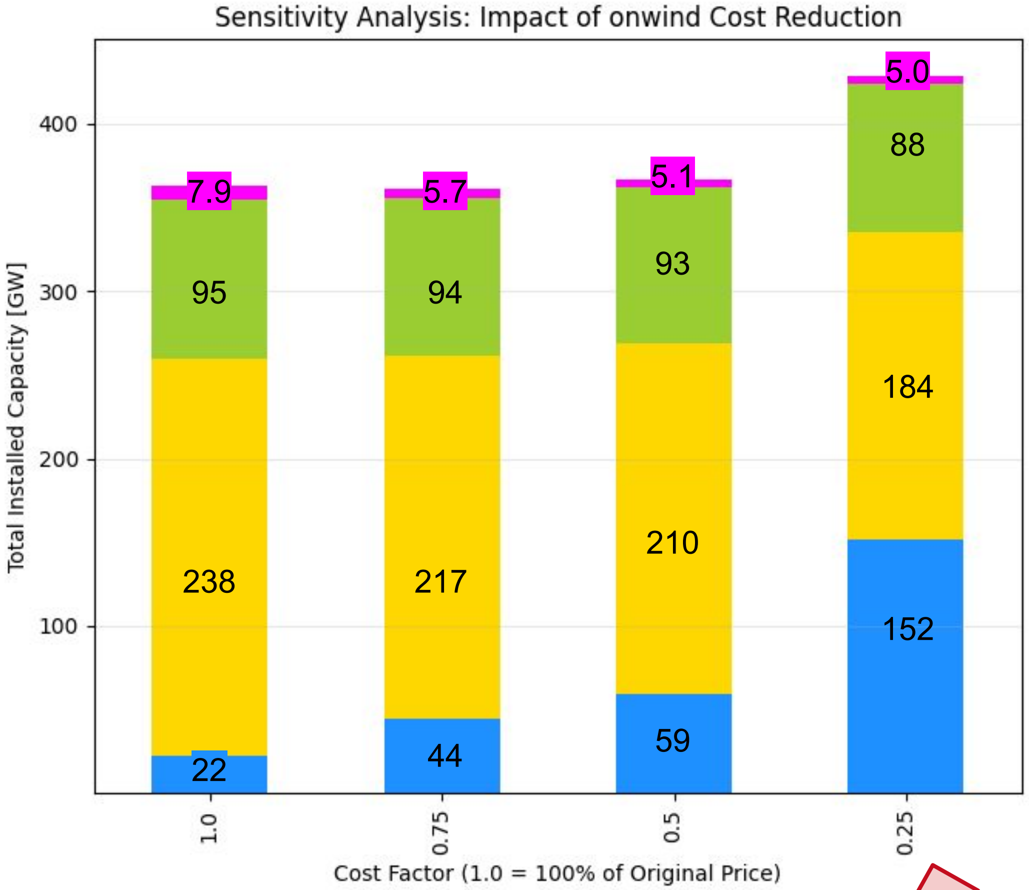
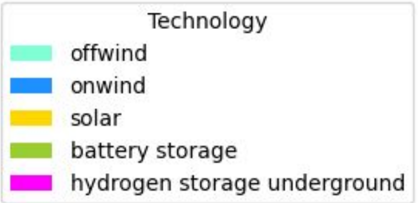
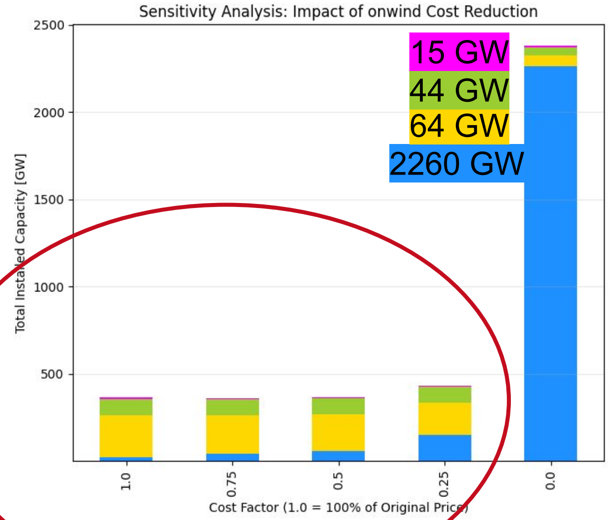
Sensitivity Analysis - Variation of Onshore Wind Capital Cost







- Lower onwind capex → onwind will replace solar & the need for storages become less, because the cheaper onwind makes curtailment more cost efficient than storages

Limitations

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Model Limitations

- **Data availability: load data (2013); Power plant data (2017); Population data (2024)**
- **Population distributed load; industry consumption not considered**
- **One bus per region & EEZ**
- Energy import/export not considered
- 2025 costs forecasted from multiple currency years at fixed discount rate
- One cost per technology (sub-type cost difference not considered)
- Storage technologies limited to H2 & BESS
- 3h time resolution
- Dispatchable technology ramping constraints not considered
- **Global fuel prices may overestimate costs for Iran (oil/gas producer)**

Improvement Potential

- Update load data to 2025
- **Allocate regional load by GDP instead of population**
- Update existing power plant data (Global Energy Monitor)
- **Use grid-based network instead of one bus per region**
- Include imports and exports with neighboring countries
- **Further subdivide EEZ regions (beyond Caspian & Persian Gulf)**
- Ensure consistent currency year for costs
- Differentiate gas, oil, and nuclear technologies
- Extend storage technologies beyond batteries and hydrogen tanks
- Increase temporal resolution from 3-hourly to hourly
- **Add technology-specific ramping constraints and costs**
- Consider domestic extraction costs and export opportunity costs

Thanks. Any Question?

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External Dataset References

(Bundesamt, 2024)

“Bevölkerung Im Jahr 2024 Um 100 000 Menschen Gewachsen.” *Statistisches Bundesamt*, 2025, www.destatis.de/DE/Presse/Pressemitteilungen/2025/01/PD25_030_124.html. Accessed 02 Feb. 2026.

(Statistical Center of Iran, 2024)

iranopendata.org/en/dataset/iod-06124-estimated-population-iran-province-2024/. Accessed 02 Feb. 2026.

(Ember Energy, 2025)

“Iran.” *Ember*, 10 Apr. 2025, ember-energy.org/countries-and-regions/iran/. Accessed 02 Feb. 2026.

(IR Gas Production, 2024)

Kgi-Admin. “Iran Natural Gas Production: Data and Insights.” *Offshore Technology*, 11 July 2024, www.offshore-technology.com/data-insights/iran-natural-gas-production/. Accessed 02 Feb. 2026.

(IR Oil Production, 2023)

LLC, Globalen. “Iran Percent of World Oil Production - Data, Chart.” *TheGlobalEconomy.Com*, www.theglobaleconomy.com/Iran/oil_production_share/. Accessed 02 Feb. 2026.

The model can be found in git repository: [Link]

Archive

The model can be found in git repository: Link

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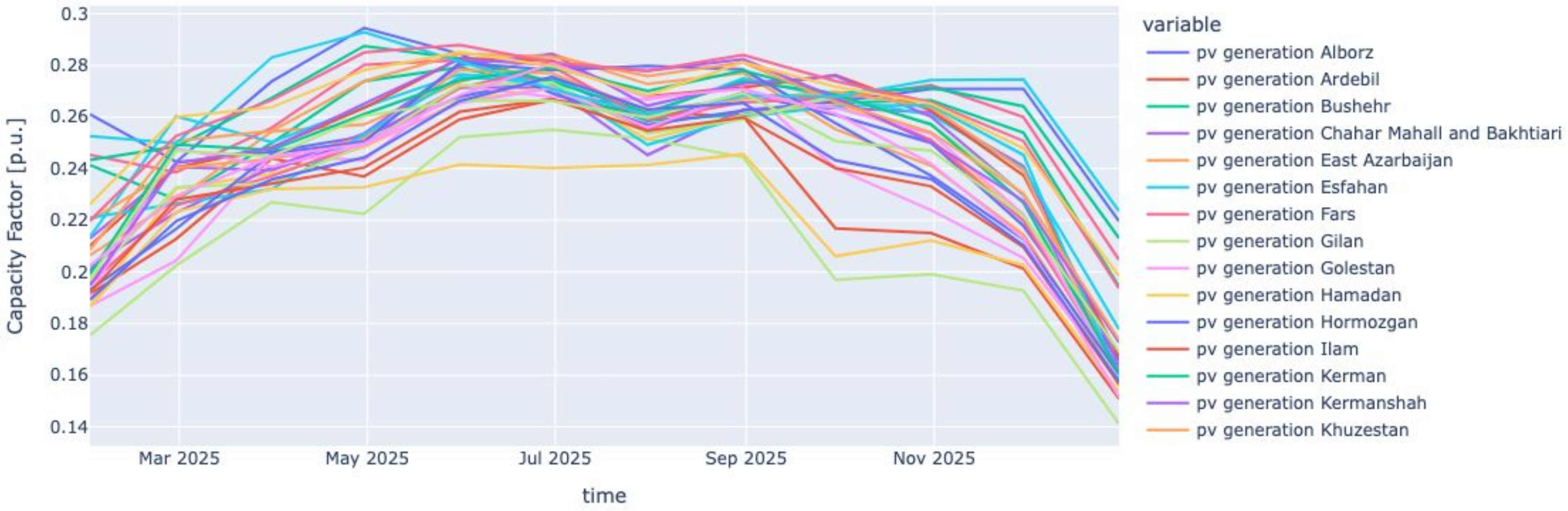
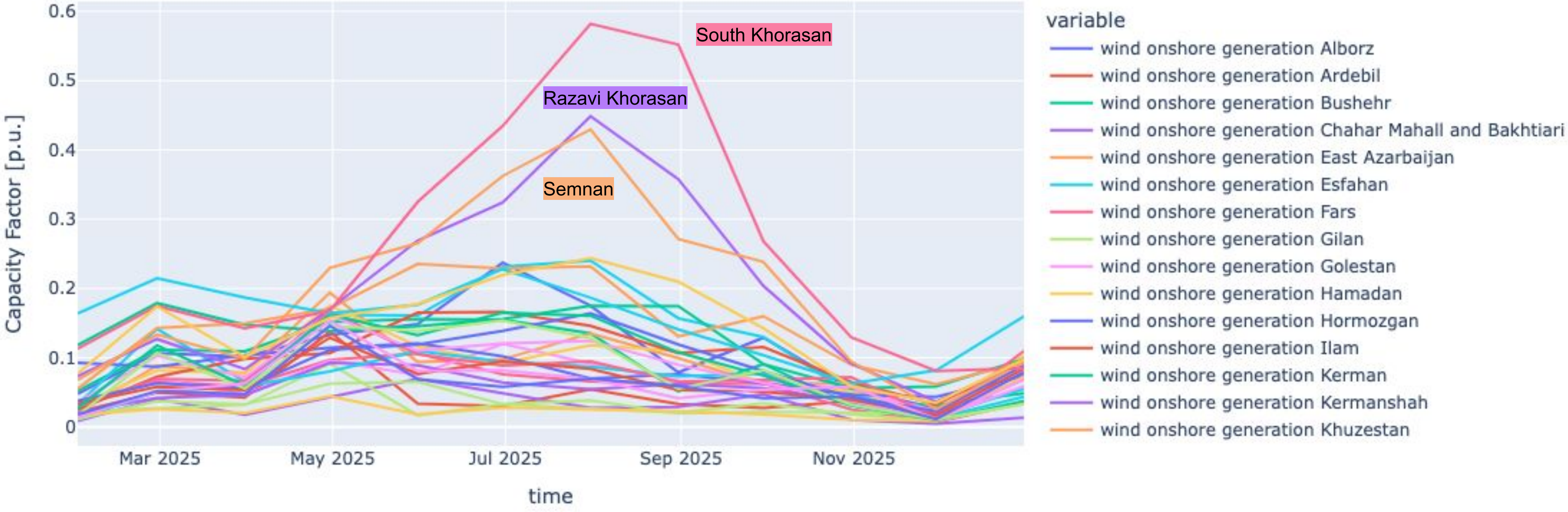
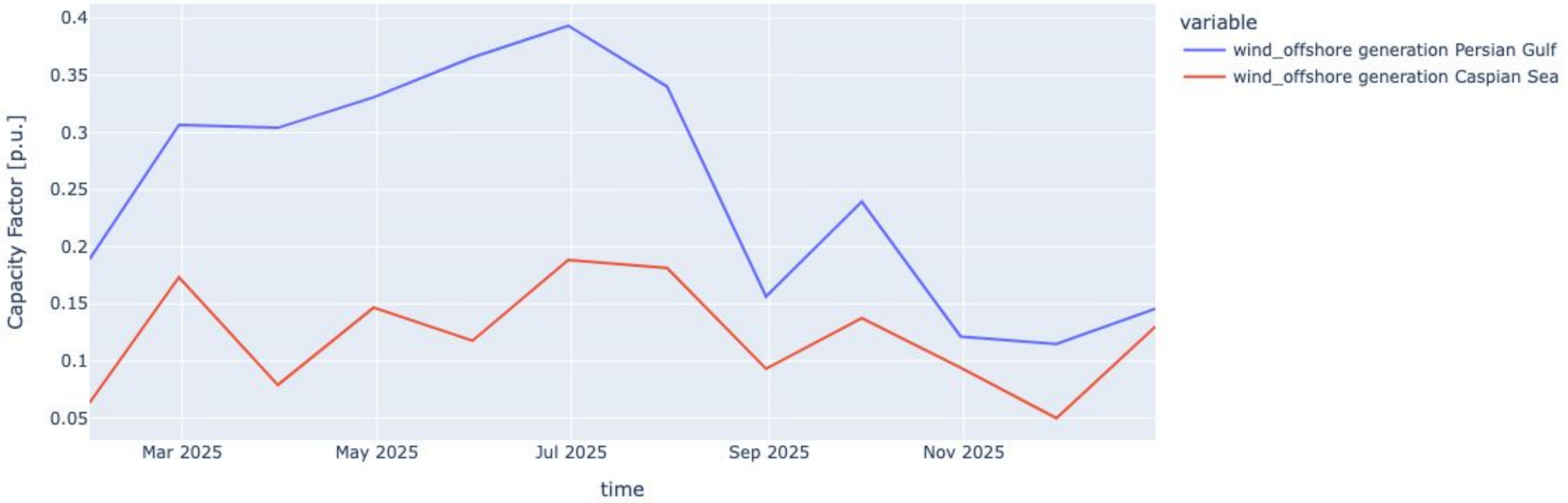
Inputs - RES Time Series (PV - Monthly Average)



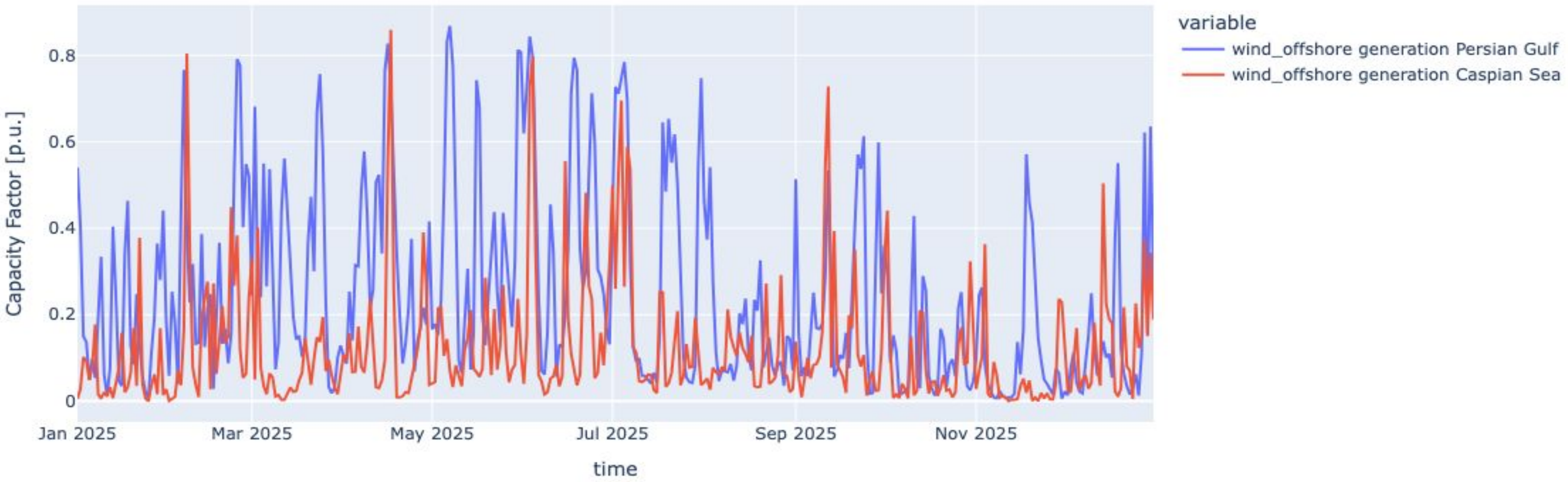
Inputs - RES Time Series (Wind onshore - Monthly Average)



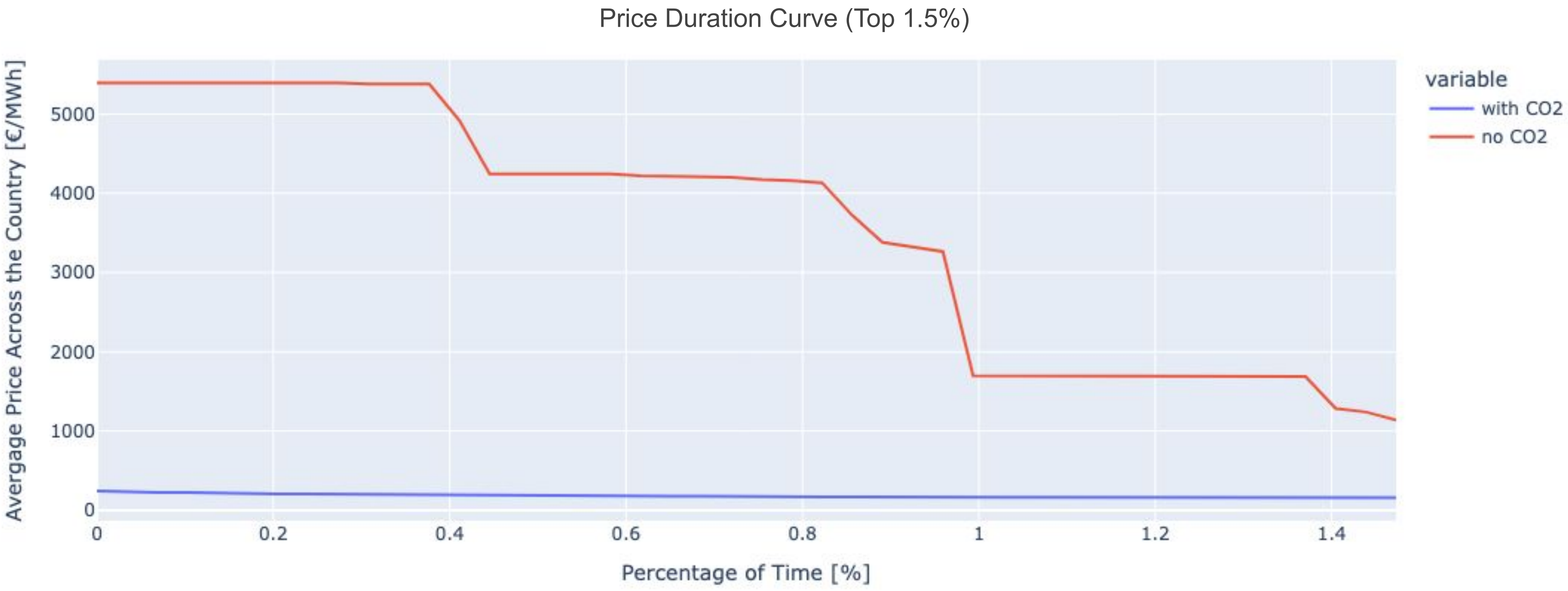
Inputs - RES Time Series (Wind offshore - Monthly Average)



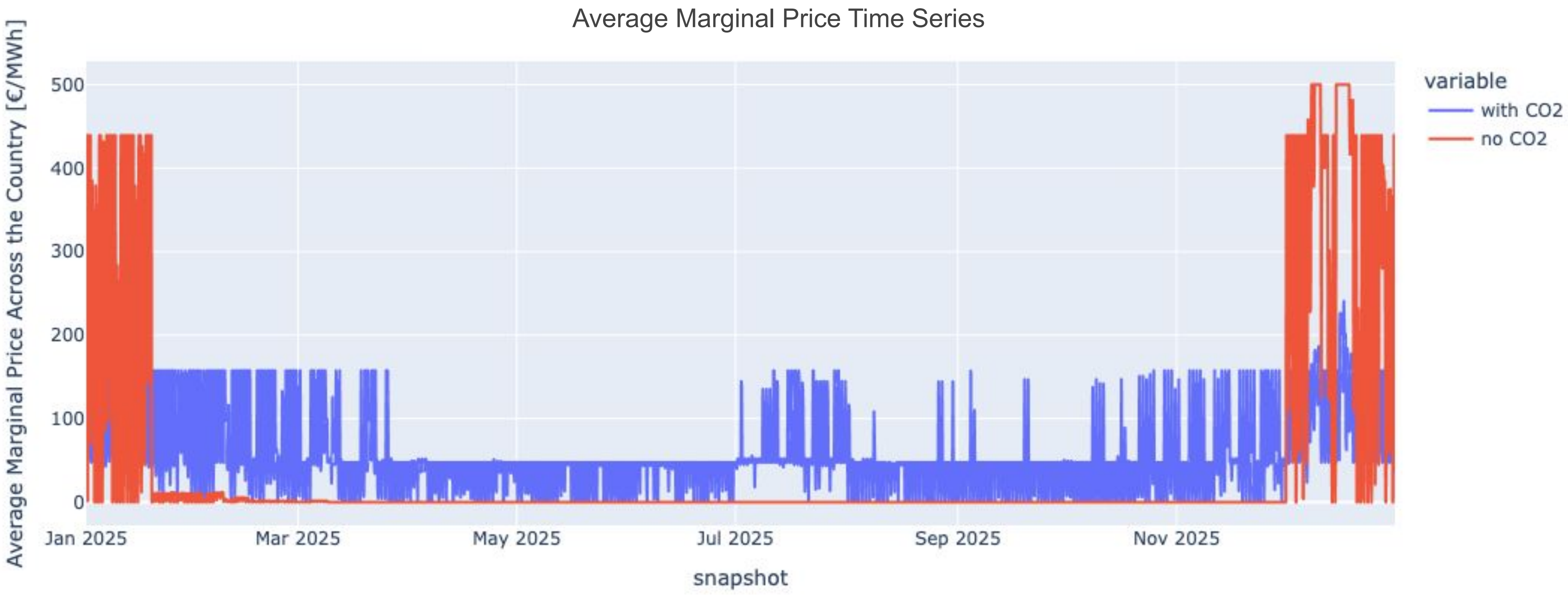
Inputs - RES Time Series (Wind offshore - Daily Average)



Results - Prices

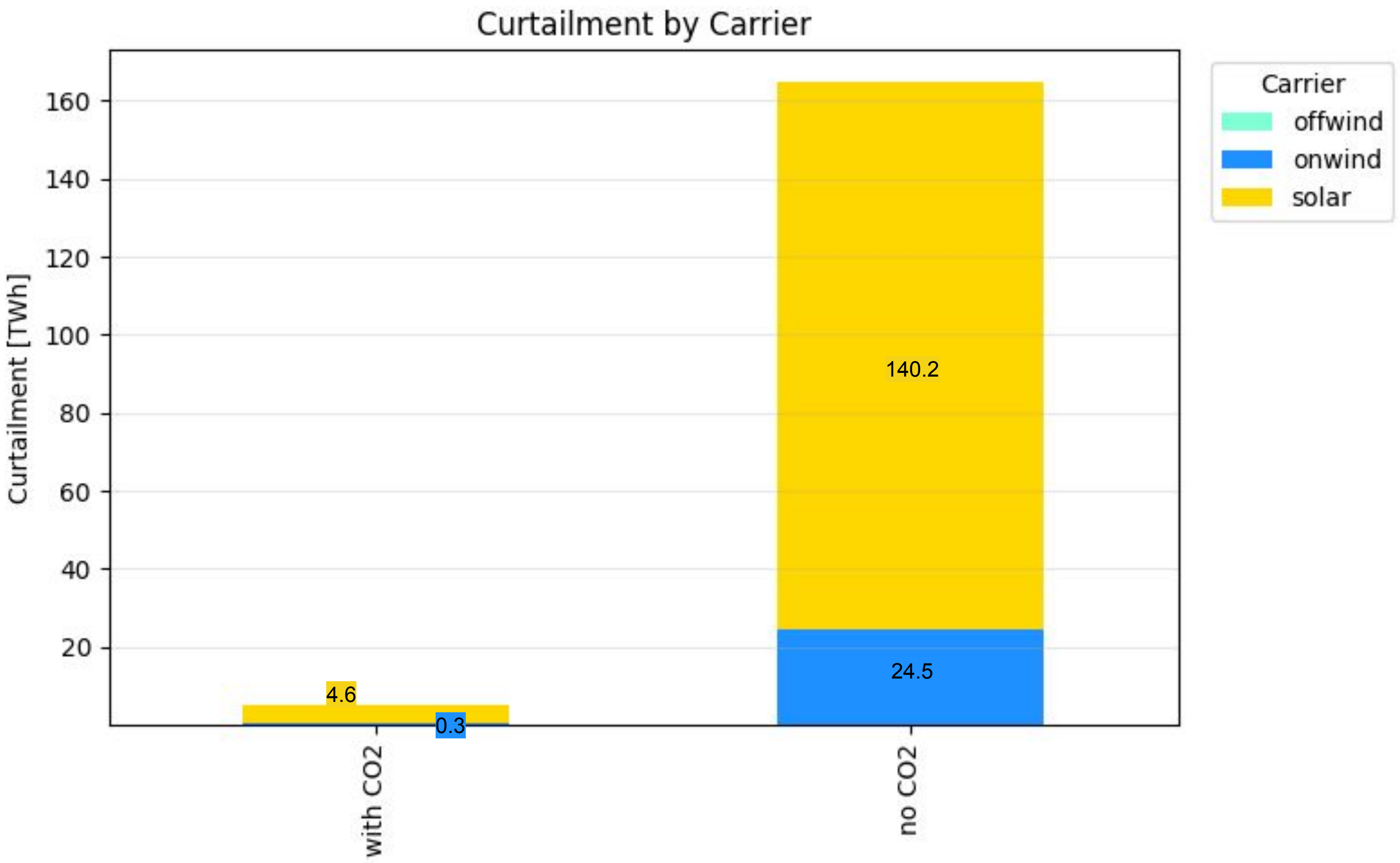


Results - Prices

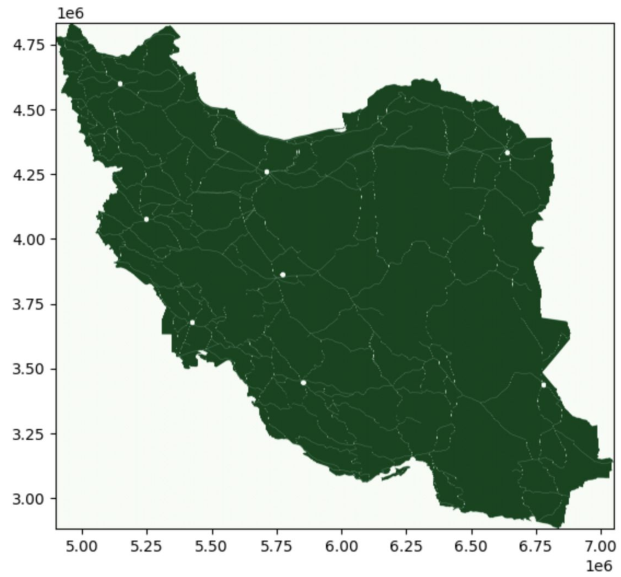


Note: Values greater than 500 €/MWh are clipped at 500/MWh for better visualization

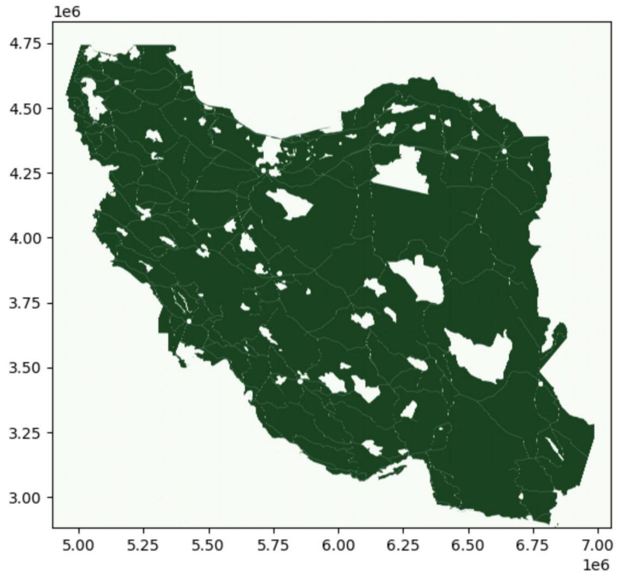
Results - Curtailed Energy



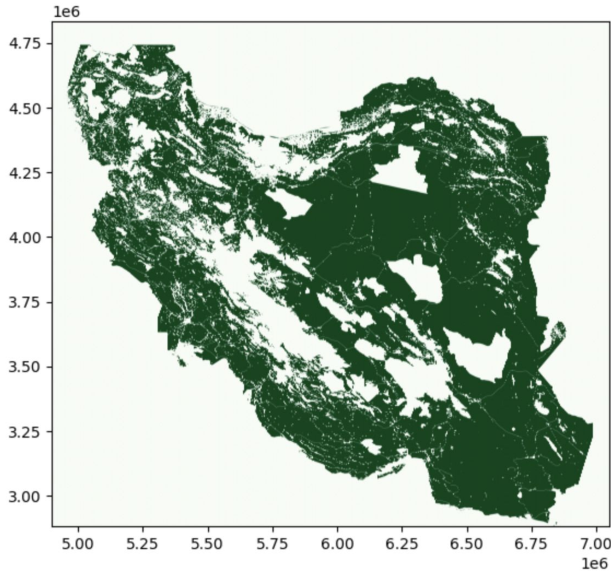
Inputs - Land Eligibility Analysis - Wind Onshore



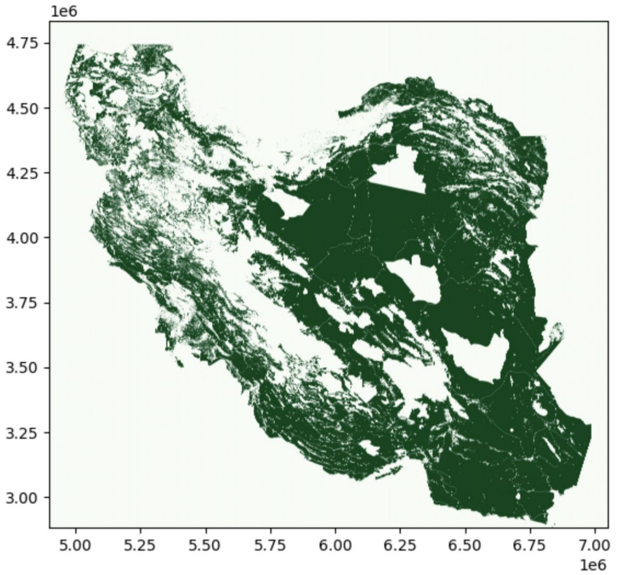
Exclusion of airports (10 km distance) & Roads (300 m distance)



Exclusion of World protected area



Exclusion of more than 2000 m Elevation & built up areas (1 km distance)

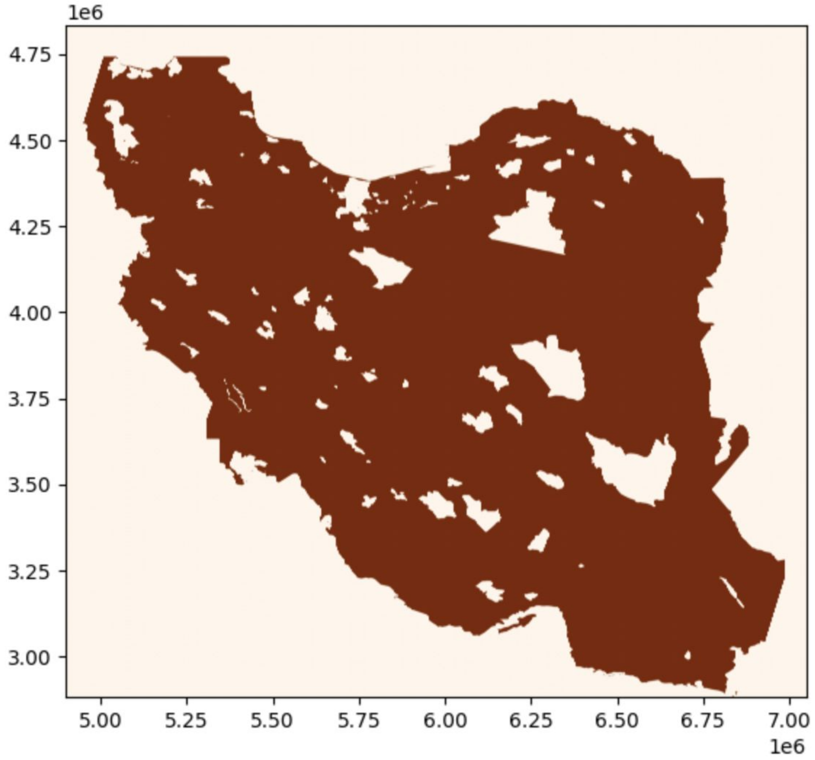


Considering the land classification

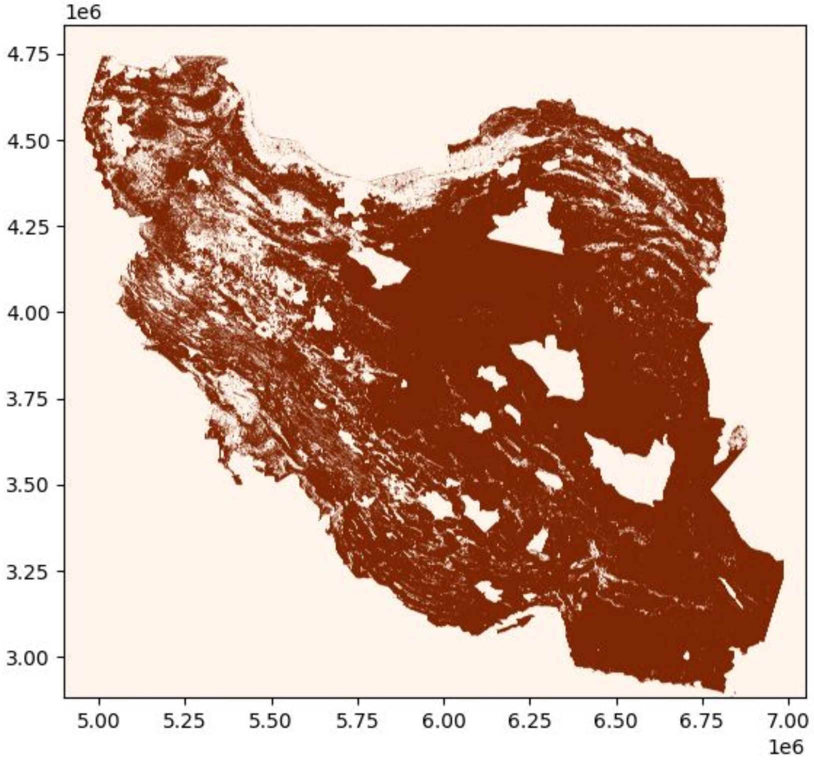
EXCLUDE: codes 100: Moss & Lichen, 90: Wetlands, 200: Open sea, 80: water bodies, 70: snow & ice, 111, 112, 113, 114, 115, 116: closed forrest

INCLUDE: codes 60: Bare / sparse vegetation, 20: Shrubs, 30: Herbaceous vegetation

Inputs - Land Eligibility Analysis - PV



Exclusion of World Protected Areas

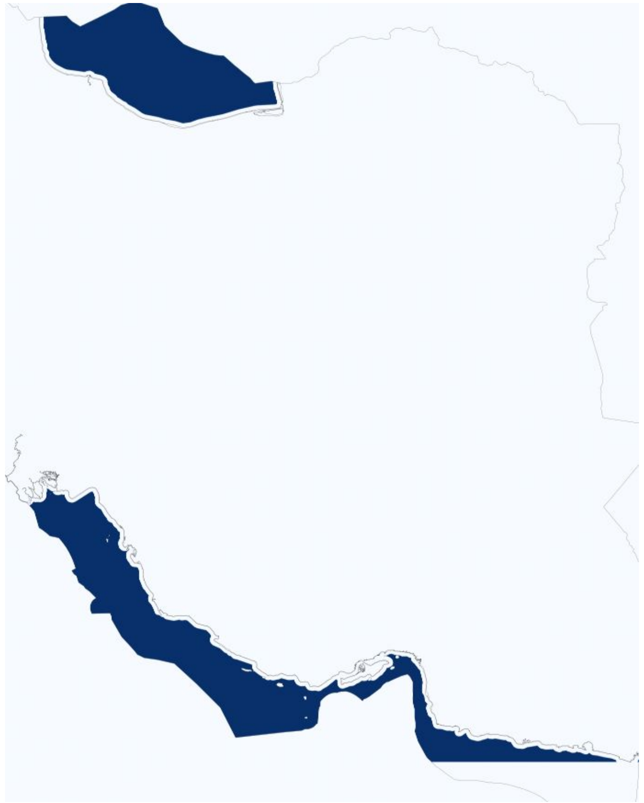


Considering the Land Classification

EXCLUDE: codes 100: Moss & Lichen, 90: Wetlands, 200: Open sea, 80: water bodies, 70: snow & ice, 111, 112, 113, 114, 115, 116: closed forrest

INCLUDE: rooftop in Built up areas (50), and utility in Shrubs (20), Herbaceous vegetation (30), Bare / sparse vegetation (60)

Inputs - Land Eligibility Analysis - Wind Offshore



Exclusion of World Protected Areas
& distance to shore (10 km)

Water depth (inclusion up to 50 m)